The Mechanism of Fields from the Bost–Connes System

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Abstract

A field is a distribution of background states and correlations formed by the joint evolution of matter and its background. Building on finite joint dynamics in the Bost–Connes state space and established matter identifications, we describe fields through source writing, background transfer, and complete reception. Actual auxiliary returns controlled by edge flux and loop exchange produce electromagnetic storage and, in a controlled long-wavelength window, the Maxwell response. Complete mass endpoints and spatial transport give massive short-range propagation and spin–isospin interaction through pion exchange; color-link loop returns give noncommuting field response. A gravitational background affects both matter maintenance phase and spatial exchange. An explicit occupation preparation with complete package reception gives distinct clock and propagation factors and realizes reciprocal clock–rod scales under finite-time error control. Source, resources, matter, and contact energy remain in one joint evolution. A separate finite construction with directional constraints supplies two accessible positive-frequency radiation modes and reciprocal absorption while retaining the full free spectrum and the unresolved material-readout connection. Each result depends on its stated preparation, spatial deployment, and working window. Relative electromagnetic calibration, persistent strong-background closure, and a common geometric identification of radiation require further dynamical relations.

Keywords: Bost–Connes system; mechanism of fields; gravitational fields; gauge fields; observed spacetime.

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§1 Definition of Fields

§1.1 Introduction

A charged body changes the motion of nearby charges; the gravity of matter deflects freely falling bodies and changes both light propagation and clock readings. These phenomena share a spatial structure: the same matter system responds differently at different positions, while different matter systems at the same position respond according to their respective properties. Comparing these responses reveals the spatial distribution of the interaction. A field describes this distribution and its variation in time.

This paper builds on the Bost–Connes system [1] and its construction of finite state-space dynamics [2]. *The Unification of Matter, Forces, and Spacetime* [3] establishes dynamical counterparts of matter and its interactions. Matter maintains itself through continual exchange with the background and interacts with other matter through that shared background. A joint action determines changes in the states of both matter and background, together with the energy and charge balances on each side. Matter that subsequently contacts this background therefore experiences a different action. The influence of one matter system on another thus follows a continuous process: an action at the source changes the background, the background transmits that change, and distant matter responds through its own coupling.

When the source persists, local exchange can maintain a stationary spatial distribution; when the source changes, the background and its action on matter evolve accordingly. Both the static form and the dynamical behavior of a field must be determined from this joint process.

The source, the background, the matter being acted upon, and the apparatus recording them all belong to the same state-space dynamics. The observational spacetime constructed in *Dynamics of the Bost–Connes State Space* provides a representation of these processes in three spatial dimensions and one temporal dimension [2]. Field positions, times, and observational outcomes use this representation; their physical content is supplied by the actual exchange, propagation, and records.

§1.2 Background Quantities of a Field

The persistent evolution of matter depends on the state of the surrounding background. Background components participating in maintenance determine the available action amplitude; components transmitting excitations carry energy, charge, and relative phase; correlations established through exchange determine how different propagation histories combine in subsequent actions. Changes that matter produces in the background affect other matter through these quantities.

Taking the background under a source-free preparation as the reference, we can track changes before and after the action of matter. For a background unit participating in maintenance, let the projector N_i represent the state in which the unit at position i is occupied and unavailable for the maintenance action under consideration. Its mean occupation is $\langle N_i \rangle$. For coherently transmitted components, the correlation $\langle B_i^\dagger B_j \rangle$ retains amplitudes and relative phases between positions or channels; here B_i is a local transition operator in the exchange being used. These quantities describe aspects of availability and coherent exchange, respectively. The

quantities required in a particular description are determined by the actual coupling between matter and background.

Let $\Omega(t)$ be the joint state containing matter, background, and the existing auxiliaries and records. Denote by $\mathcal{B}$ the background subsystem that carries the interaction under consideration in the actual joint construction. Its reduced state and a local reading over a region U can be written as

$$\rho_{\mathcal{B}}(t) = \mathrm{Tr}_{\overline{\mathcal{B}}}\,\Omega(t), \qquad b_O(U,t) = \mathrm{Tr}[\rho_{\mathcal{B}}(t)O_U]\,. \tag{1.1}$$

Here O_U is a background observable acting on that region, with regions defined by the actual position degrees of freedom. Correlations across regions are computed using the corresponding joint observables. Correlations already established between background and matter remain in $\Omega(t)$ and continue to participate in subsequent evolution. This description accommodates local occupation, relative phase, and the correlations required for propagation.

Spatial positions follow the observational construction of Ref. [2] and correspond to positions in the actual exchange actions [2, 3]. The same type of matter may occur at different positions, and the same position may participate in exchanges through different channels. Matter type, background spectral label, and propagation position each retain their own physical meaning. Which units can exchange directly in space is determined by the specific interaction. Both local field quantities and their spatial connections therefore refer to definite objects.

A uniform reference background makes identical matter experience the same local action everywhere. After a matter source changes part of the background, the available action and exchange states differ from position to position. Propagation develops these differences into a spatial distribution, which constitutes the field.

§1.3 Field Formation

A field source arises through actual contact between matter and background. The persistent action of matter depends on the background, and the background receives the influence of matter in that same action. The resulting change may be a change in the occupation of background units, the injection of energy-carrying or charged excitations, or a change in relative phases and correlations between propagating components. The next contact begins from these altered states, allowing the source action to accumulate through successive evolution.

Reference [3] gives a source process that retains newly produced outputs. The local generator of matter acts on a freshly prepared two-state auxiliary, writing its action strength into the auxiliary output. This output then exchanges with the local background unit, while the old output remains in the joint system. Let $\xi_i = \langle N_i \rangle - 1/2$ denote the occupation change relative to an unbiased preparation, and let ε_i be the probability of executing this complete update. We use $\hbar = 1$. For the generator eigenvalue λ_i of a fixed source at rest, this realization gives

$$\xi_i' = (1-\varepsilon_i)\xi_i + \varepsilon_i b_i, \qquad b_i = \frac{1}{2}\sin(2t_{\mathrm{read}}\lambda_i). \tag{1.2}$$

Here t_{read} is the effective interaction duration between source and auxiliary. When the update is executed, the local unit acquires the new occupation produced by the source; otherwise its previous state is retained. Source writing and background

replenishment jointly determine the net change in each cycle. As the occupation approaches the value corresponding to the source output, the difference available for further writing decreases. Exchange with the surroundings can carry this bias away, allowing the source to continue producing a net output into the background.

Coherent exchange retains another kind of output. During emission, transmission, absorption, and retrieval between matter systems, a component leaving the emitting matter reaches the receiving matter through the same propagation process. The final auxiliary state and propagation phase jointly determine the final amplitude [3]. Retrieval can restore some auxiliaries to their prepared states while leaving phases or correlations in the matter at the two ends. The background description required for a persistent interaction therefore includes components still participating in propagation and return, as well as their correlations with matter.

These actions can all be composed cycle by cycle within the full joint evolution, using a joint description of repeated interactions [4]. If the auxiliary newly added in cycle n is prepared in η_n, and the actual joint unitary action in that cycle is $\mathbb{U}_n$, then

$$\Omega_{n+1} = \mathbb{U}_n(\Omega_n \otimes \eta_n)\mathbb{U}_n^\dagger. \tag{1.3}$$

The existing background, matter, and records remain in Ω_n; the new auxiliary is prepared according to the specified process. The actual ordering of source action, propagation, and return is contained in $\mathbb{U}_n$. Reduction to the background degrees of freedom gives the background state left by the cycle, while reduction to matter gives the change in the source itself. The field source and the matter response thus come from the same joint final state.

A persistent source repeats this process. Whether a field can be maintained, how long it takes to build up, and how long it persists after the source changes are determined by the joint evolution of writing, propagation, replenishment, and return. The preparation maintaining the source state also belongs to this process; the required supply and retrieval are included in the physical balance.

§1.4 Field Propagation

A background change produced at the source can affect another position only through actual exchange. A local contact transfers the change to background units coupled to the source. These units subsequently exchange with the background at other positions, or propagate while carrying excitations, and then contact distant matter. Each segment of the action retains its state, phase, and elapsed duration. Successive actions extend a local source into a spatial response.

In the background-update realization of Eq. (1.2), background units subsequently undergo the specified exchanges. If R is the transport matrix for the first moments of occupation, and $D_\varepsilon = \mathrm{diag}(\varepsilon_i)$, the process of updating first and transporting afterward gives [3]

$$\boldsymbol{\xi}_{n+1} = R(I - D_\varepsilon)\boldsymbol{\xi}_n + RD_\varepsilon \boldsymbol{b}_n. \tag{1.4}$$

This equation directly connects the source, propagation, and the distribution already formed: the first term transports the retained background bias, and the second transports the bias newly written in the current cycle. After the source action, background returning to the source region enters the next writing step, so the net

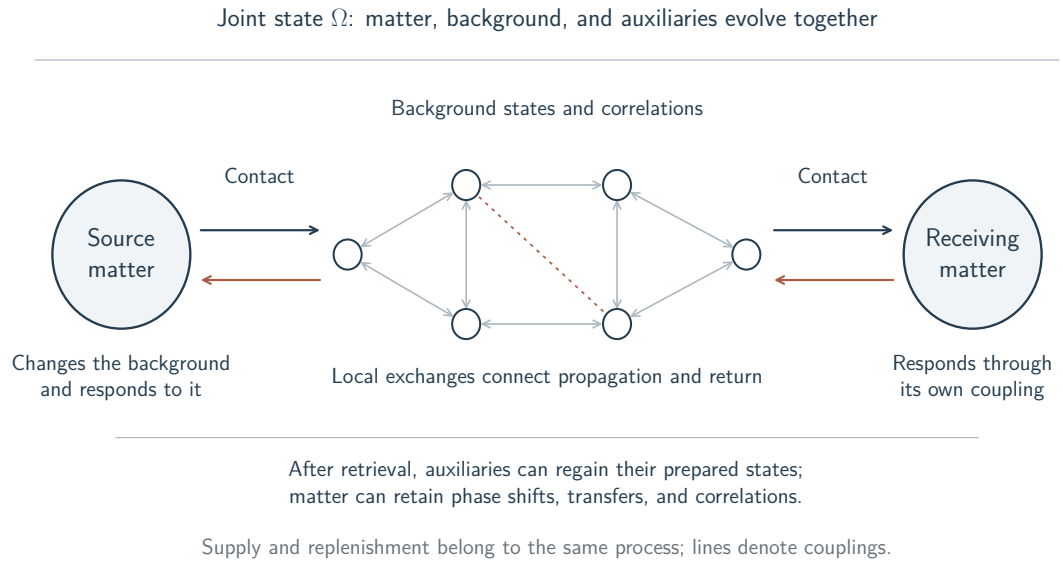


Figure 1.1. Joint evolution of matter and background. Contact with the source changes the background, and local exchange transmits this change to other matter. Back-reaction and return on all sides remain within the same joint state. Nodes and links represent background units and their couplings; the dashed line indicates correlations that must be retained. When retrieval restores an auxiliary to its prepared state, matter may still retain the phase changes, transfers, and correlations produced by the action.

source output adjusts together with the surrounding distribution. This occupation recurrence follows from the full exchanges and repreparations used here; coherent propagation is calculated from the corresponding amplitudes and correlations in Eq. (1.3).

For a fixed source, if this update converges under the supply boundary conditions, it ultimately forms a stationary distribution at the corresponding phase of the cycle. Local writing, transport, and replenishment still occur in every cycle, with the changes balancing at the end of each cycle. A static field is maintained by ongoing dynamics. When the source changes in time, new outputs enter the same exchange process in succession, and the previous distribution gradually becomes a new one. Field build-up and variation follow from the same evolution.

The spatial form depends on the actual propagation operator and boundary conditions. For example, equally weighted nearest-neighbor exchange in three dimensions gives a discrete harmonic equation outside the source, with a $1/r$ distribution in its controlled far-field regime [3]. This field profile follows from conservative exchange in three-dimensional space and the corresponding low-wavenumber response. Changing propagation, restoration, or the boundary changes both the spatial distribution and the build-up time. Source strength and spatial field profile are related within one process, while their respective origins remain distinct.

Propagation time accumulates over the complete process: contact, preparation, transport, retrieval, and waiting all contribute to the actual elapsed duration. The number of exchanges needed to cross a region and the duration of each exchange determine the propagation response. The spatial distribution of a stationary field, the arrival of a time-dependent response, and the speed of phase propagation are

each read from this actual evolution.

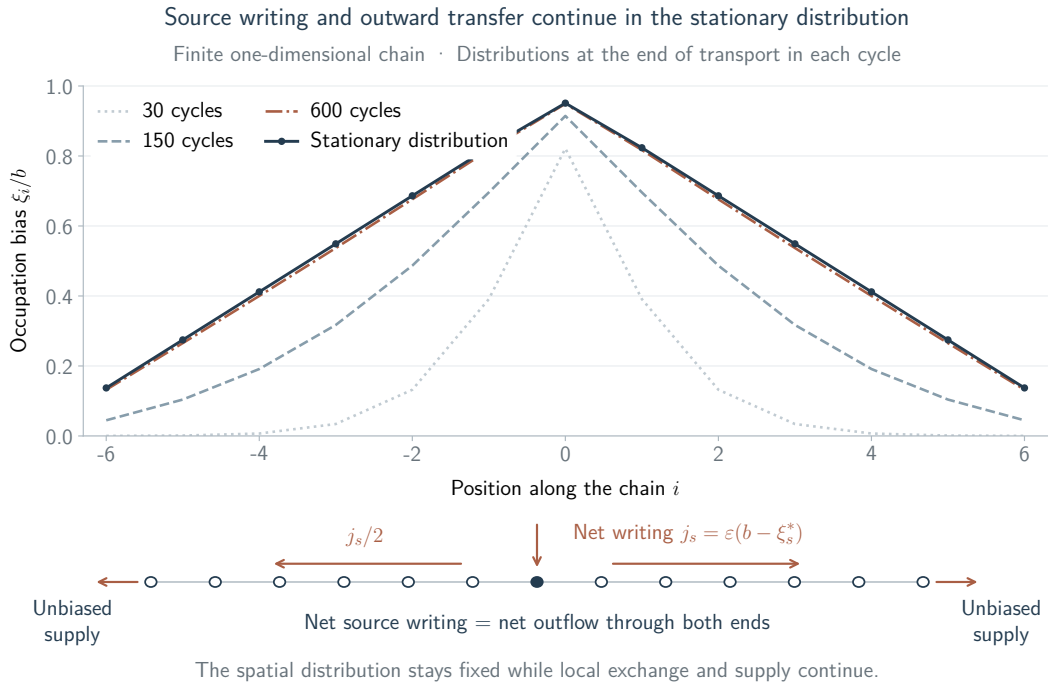


Figure 1.2. Dynamical maintenance of a stationary distribution. A finite chain of thirteen units governed by Eq. (1.4) illustrates the build-up of the distribution. The central source is continually updated, and both ends exchange with an unbiased supply. All curves show occupations at the end of transport in each cycle. In the stationary state, net source writing is exactly balanced by net outflow through the two ends, while exchange continues. Here $\varepsilon = 0.2$ and $b = 0.2$; at most one edge is selected for exchange per cycle, and each internal or supply edge has selection probability $1/28$. These parameters serve only to illustrate the recurrence mechanism. This is a one-dimensional example; the $1/r$ far field discussed in the text corresponds to three-dimensional propagation.

§1.5 Field Action

After a background change reaches a distant position, it enters the evolution of local matter through that matter's existing coupling. Available background states change the amplitude of the persistent action of matter, while transmitted excitations and phases alter exchange through compatible channels. Source matter and receiving matter are thereby connected through the same background. Treating one matter system as a probe means using its response to describe this interaction.

Field action can be determined by comparison with a reference process. Fix the probe preparation, calibration scales, and actual elapsed duration T. Let $\rho_{\mathrm{T}}^{(s)}(T)$ and $\rho_{\mathrm{T}}^{(0)}(T)$ denote the probe outputs with the source action under consideration present and with that action switched off, respectively. For a calibrated probe observable O_{T}, the source-induced change is

$$\delta\langle O_{\mathrm{T}}\rangle = \mathrm{Tr}\Big[O_{\mathrm{T}}\big(\rho_{\mathrm{T}}^{(s)}(T) - \rho_{\mathrm{T}}^{(0)}(T)\big)\Big]\,. \tag{1.5}$$

Both outputs are calculated from actual processes containing the source, background, and probe. Choosing position, coherence, or momentum observables gives

changes in distribution, changes in phase and visibility, or momentum transfer, respectively. The momentum difference divided by the total elapsed duration of the action gives the mean force relative to the reference process [3]. The mass and charges carried by the probe determine its coupling. When probe properties can be factored out of the response, the remaining spatial and temporal dependence gives the physical quantities of that type of field.

Coherent action can also directly connect phase and motion. When the actual output in the regime considered is described by an effective generator $H_{\mathrm{T}} = H_{\mathrm{T},0} + \delta H_{\mathrm{T}}$, a background change makes matter accumulate a different phase. For a fixed eigenbranch that remains occupied, let $\delta h(t)$ be the change in its generator eigenvalue. Then

$$\delta\varphi = -\int_0^T \delta h(t)\, dt, \qquad \frac{d\langle P\rangle}{dt} = i\langle[H_{\mathrm{T}}, P]\rangle. \tag{1.6}$$

Here P is a calibrated momentum operator with no explicit time dependence. The first expression gives the dynamical phase accumulated along the actual evolution; the second gives the momentum change caused by the same action. In a continuum position representation, if the generator change takes the form of a local potential energy $V(x,t)$, its contribution to the rate of momentum change is $-\langle\nabla V\rangle$. At discrete positions, the corresponding commutator is retained. Spatial differences in the field become forces through matter evolution; phase and impulse are different readings of this action.

The probe generally also affects the background. Treating the source as fixed and the probe as passive is a description valid within a regime of the full process. When back-reaction is required, source, background, and probe continue to evolve jointly. Joint conservation relations in the same realization account for energy and charge changes on each side [2]. This balance connects field action to field formation and supplies a basis for the subsequent discussion of time-dependent fields and radiation.

§1.6 Field Observation and Definition

Field action leaves records in matter and apparatus. A moving probe changes its position distribution, an interferometer converts relative phase into output differences, and a clock reflects the local rhythm of evolution by comparing successive processes. The corresponding distinctions are stored in the probe and actual auxiliary degrees of freedom and are read out as events according to the recording scheme. Field observation thus occurs within the same state-space dynamics that produces the field.

Actual pointers determine record components [2]; outcome probabilities and conditional final states together describe a measurement instrument [5]. Let Ω^{out} be the joint final state including detection, and let Π_r project onto pointer outcome r. Then

$$p_r = \mathrm{Tr}\big[(I \otimes \Pi_r)\Omega^{\mathrm{out}}\big], \qquad \sum_r p_r = 1. \tag{1.7}$$

The probabilities of the different outcomes and their conditional final states are supplied by the same evolution, and the state left after readout continues into subsequent processes. Expressing event positions, times, and calibrated responses

in a common $3+1$ representation allows probe responses in different regions to be compared. Background propagation histories, the forces acting on matter, and the final events are linked in sequence within this representation.

Each kind of observation retains the background information it requires. Occupation changes the available action amplitude, relative phase affects interference, and correlations between background and matter or between regions affect repeated contacts. Identical local occupations can correspond to different phase responses. A field description therefore retains the states and correlations relevant to the action under study, and a specific apparatus extracts measurable field quantities from them.

A field can accordingly be defined as **the distribution of background states and correlations over position and time in an actual interaction; its influence on matter is determined by responses and records within the same evolution.** This paper mainly concerns changes produced and maintained by matter sources in a reference background.

Events collected over space and time give statistical field quantities. In an observational regime with stationary responses, finite correlation times, and sufficiently slow field variation, regional averages can stably characterize the local field. Smooth spatial variation further allows discrete relations to be written as continuum field equations. Statistical precision is determined by the actual event history and its correlations, while the spatial approximation depends on the relation between propagation and observation scales. The continuum field description is thereby connected to the underlying discrete process.

Field and force have related but distinct meanings here: a field describes the background state and distribution through which an interaction occurs, while force describes the momentum change of matter under that action.

§1.7 Types of Fields

Changes that matter produces in the background can affect other matter through different modes of action. Dynamical counterparts of gravity and gauge interactions are established in Ref. [3]. We retain these physical objects and develop their field mechanisms through their sources, propagation, and responses.

A gravitational field involves modulation of the persistent evolution of matter by the shared background. The source changes the available background states; propagation distributes this change to other positions, where matter experiences it through the complete maintenance action. Changes in phase, timing, and motion associated with the common modulation provide physical readings of gravitational potential, spacetime calibration, and gravitational action. Their spatial distribution is determined by the joint response of source and background.

Gauge fields involve exchanges between matter systems through compatible channels. The source charges and exchange structure determine which components can be emitted, propagated, and received. Relative phases and multichannel correlations determine subsequent actions. Different charged matter systems respond to the same propagation process according to their respective couplings. Electromagnetic, weak, and color interactions are developed using their established physical counterparts; nuclear forces are discussed through collective exchange between composite matter systems.

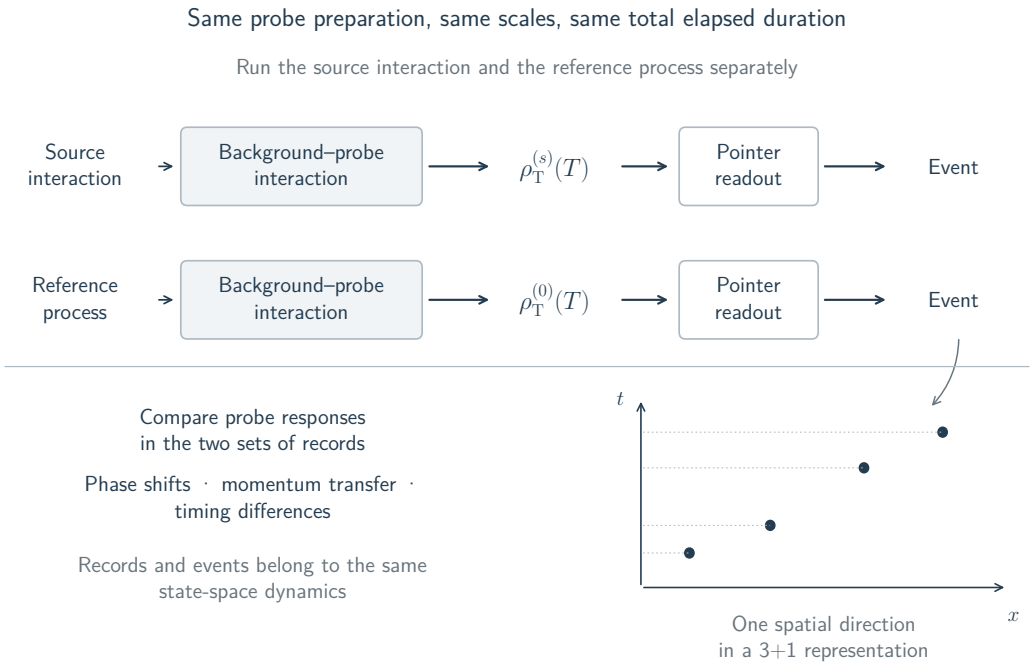


Figure 1.3. From probe response to observational events. The process with source action and the reference process evolve separately, and their records are compared using the same probe preparation, calibration scales, and total elapsed duration. Pointer coupling leaves records of the probe output, and reading the outcomes constitutes events. Event positions and times are represented in the same three spatial dimensions and one temporal dimension. The lower-right panel shows only one spatial direction and time. Points denote selected events, without implying that every evolution cycle produces a record. Comparing the two sets of processes determines the response difference.

The two types of fields differ in the background quantities changed by the source, their propagation structures, and their couplings to matter. The equations and physical properties of each field are determined as these processes are developed.

§2 Gravitational Fields

A gravitational field changes the rhythm of matter evolution at different positions. A matter source changes the available share of the nearby background, and background exchange transmits this change elsewhere. Local matter continues to accumulate phase through the same maintenance action and therefore acquires position-dependent energy and clock rates. Spatial gradients deflect its motion, while time calibration makes clocks at different positions accumulate different readings. Potential, force, and timing arise from this common response.

Chapter 1 connected source writing, background propagation, and matter response. Here we first construct a static gravitational field using the continual-refreshing process of Ref. [3], adopting the averaged, long-wavelength, and weak-field description of motion already obtained for that process. We then discuss the relation of the spatial length scale to actual propagation, the general transport structure, and the time-dependent response. Time is accumulated over the complete process, and energy and length use the same distant reference. Denote the speed of light in this reference by c_0; below we set $\hbar = c_0 = 1$.

§2.1 Gravitational Potential

The phase of matter is determined by the maintenance generator. When background availability decreases, this action weakens, and phase accumulation slows relative to the reference background. For a single two-state resource unit in Chapter 1, the available response is $h_i = 1 - N_i$, with mean reference response $h_0 = 1/2$ under the unbiased preparation. Once the source establishes a stationary bias, the mean response and gravitational potential are

$$\bar{h}_i = h_0 - \xi_{*,i}, \qquad \Phi_i = \frac{\bar{h}_i - h_0}{h_0} = -2\xi_{*,i}. \tag{2.1}$$

In the positive-bias source regime used here, the source increases the unavailable share of the background, so $\xi_{*,i} > 0$ and $\Phi_i < 0$. The distant unbiased supply gives $\Phi \to 0$ and fixes the reference potential. The potential has a concrete physical meaning: it is the change in the local maintenance response relative to the common baseline.

Let m be the rest energy of test matter without source action. Over the same complete cycle, the mean frequency of the maintenance phase changes from m to $m(1 + \Phi_i)$, so the rest-energy change is

$$V_i = m\Phi_i. \tag{2.2}$$

Matter type enters the potential energy through m, while the spatial variation of the background is described by Φ_i. Under the same common maintenance modulation, different matter systems read the same potential. They still have their own masses, internal structures, and channel interactions; the gravitational response acts commonly on their persistent evolution.

The occupation–mean-response relation in Eq. (2.1) is exact for this resource realization. Equation (2.2) applies it within the controlled regime of motion in Ref. [3].

§2.2 Sources and Spatial Distributions

Net writing at the source in each cycle determines how much bias the background must transport outward. Let $\boldsymbol{\xi}^+$ be the stationary distribution after refreshing and before transport, let c be the selection probability of each internal exchange edge and each supply edge, and let L be the corresponding graph Laplacian. The update of Chapter 1 gives

$$cL\boldsymbol{\xi}^+ = \boldsymbol{j}, \qquad \boldsymbol{j} = D_\varepsilon(\boldsymbol{b} - \boldsymbol{\xi}_*), \qquad \boldsymbol{\xi}_* = (I - cL)\boldsymbol{\xi}^+. \tag{2.3}$$

Here $\boldsymbol{j}$ is the actual net source. The bias already accumulated in the background enters the next update and reduces the difference still available for source writing. The net source thus depends both on the source itself and on the returning background. Where there is no refreshing outside the source, local exchange only transfers existing bias, and Eq. (2.3) becomes a discrete harmonic equation [6]. Summing over all positions cancels internal exchanges pairwise, equating net source writing with net outflow through the supply boundary. A stationary distribution is maintained by a continuing flow.

For a single source at s, write $\mathcal{G} = L^{-1}$ for the Green kernel with the given supply boundary conditions. With source refreshing probability ε, eliminating the source occupation from Eq. (2.3) gives [3]

$$j_s = \frac{c\varepsilon b_s}{c(1-\varepsilon) + \varepsilon\mathcal{G}_{ss}}, \qquad \xi_{*,i} = \frac{j_s}{c}\mathcal{G}_{is} \quad (i \neq s). \tag{2.4}$$

The kernel $\mathcal{G}_{is}$ gives the response from the source to position i, while $\mathcal{G}_{ss}$ enters the source's own feedback. The distant amplitude and local net writing are determined by the same spatial process. The inverse exists under the given boundary conditions when every relevant connected component can reach the supply. A corresponding large-volume limit can then be taken for regions far from the boundary.

Take actual positions on a three-dimensional nearest-neighbor grid with equal step lengths and equal exchange weights in all six directions. Denoting the step length by ℓ_x, the low-wavenumber part satisfies $L \simeq -\ell_x^2\nabla^2$. In the far region outside the source, the kernel is

$$\mathcal{G}_{is} \simeq \frac{\ell_x}{4\pi r}, \qquad \Phi(r) \simeq -\frac{j_s\ell_x}{4\pi c h_0}\frac{1}{r}, \qquad r = |\boldsymbol{x}_i - \boldsymbol{x}_s|. \tag{2.5}$$

This $1/r$ dependence follows from the low-wavenumber response of three-dimensional conservative exchange. Source writing fixes the overall amplitude, while propagation in three dimensions fixes the radial decay. Finite lattice spacing, distance to the boundary, and the spatial extent of the source determine departures from the far-field expression. The detailed profile remains calculable from Eq. (2.3).

For a weak source, $b_s = t_{\mathrm{read}}\lambda_s + O((t_{\mathrm{read}}\lambda_s)^3)$. Calibrating λ_s by the source's own rest energy M_s measured with the same background clock makes net writing proportional to M_s at leading order. For distributions to which the common source calibration applies and for which feedback between sources is negligible, the leading-order continuum description is

$$\boxed{\nabla^2\Phi = 4\pi G\rho_M, \qquad \Phi(r) = -\frac{GM_s}{r}.} \tag{2.6}$$

Here ρ_M is the mass density of matter sources at rest; the point-source expression applies in their exterior far field. The mass of a composite source is calibrated by the total generator actually maintained, with binding contributions included in the total phase. Replenishment at the source, feedback between neighboring sources, and finite-source corrections remain in the net writing of Eq. (2.3). Source measures for moving matter and freely propagating radiation are discussed in Sections 4.2–4.3.

§2.3 Newton's Constant

Newton's constant compares two outputs of the same process: the mass phase accumulated by the source, and the potential energy acquired by distant matter through the source-induced background change. Equation (2.5) directly gives the leading strength of the process used here,

$$G_{\mathrm{dyn}}^{(0)} = \frac{\ell_x}{4\pi c h_0} \lim_{M_s \to 0} \frac{j_s}{M_s}. \tag{2.7}$$

The source-readout duration, exchange selection in each cycle, return feedback, and complete timing all enter this quantity through j_s/M_s, c, and ℓ_x. They determine the amplitude of the actual output and cannot be eliminated from the $1/r$ profile alone.

The common physical units follow the background spectral calibration of Ref. [3]. If W_{met} is the effective participation weight of the gravitational process, C_{bg} is the total squared spectral gradient of the background used, and E_0 is the energy unit, then

$$E_{\mathrm{met}}^2 C_{\mathrm{bg}} = E_0^2 W_{\mathrm{met}} (\log W_{\mathrm{met}})^2, \qquad G_E = E_{\mathrm{met}}^{-2}. \tag{2.8}$$

We retain the single physical calibration of Ref. [3]: $G_{\mathrm{dyn}}^{(0)} = G_E$ fixes the correspondence between the gravitational process used here and the spectral unit. The common calibrated strength is denoted by G in the field equations. This matches the actual strength to the common energy scale; the details of source and return remain available for computing finite corrections.

At the participation depth and energy unit adopted in Ref. [3], the spectral scale gives $G_E \simeq 6.70272 \times 10^{-39}\,\mathrm{GeV}^{-2}$.

§2.4 Matter Motion and Clock Rates

The same maintenance response enters both rest energy and the energy relation for motion. The passive motion realization in Ref. [3] yields a positive-energy branch when the local background varies slowly. In an isotropic long-wavelength, low-speed representation at the same three-dimensional positions, with $c_0 = 1$, its leading expansion is

$$E_+(x,p) = m + m\Phi(x) + \frac{|p|^2}{2m} + \cdots . \tag{2.9}$$

The rest and kinetic terms use the same m. The wave-packet center moves according to spatial phase variations, hence

$$\dot{\boldsymbol{X}} = \frac{\boldsymbol{p}}{m}, \qquad \dot{\boldsymbol{p}} = -m\nabla\Phi, \qquad \boxed{\ddot{\boldsymbol{X}} = -\nabla\Phi.} \tag{2.10}$$

The test mass cancels from the leading acceleration because the inertial and potential-energy coefficients belong to the same process. For a point source, $\Phi = -GM_s/r$ increases with r; its negative gradient points toward the source and thus gives attraction. Higher-order motion retains the effects of lattice dispersion, background gradients, and resource fluctuations. Momentum changes in a finite process are calculated from the actual generator of Chapter 1.

Clock rates read the same matter phase. Let $A(x)$ be the ratio of the local clock frequency to that of the distant reference. For the common response above,

$$A(x) = 1 + \Phi(x), \qquad d\tau_{\mathrm{loc}} = A(x)\, dt. \tag{2.11}$$

Here τ_{loc} is defined by phase counting of the local clock, while t uses the distant reference. Near the source, $A < 1$, so less clock phase accumulates during the same reference duration. Clocks at different positions can be compared by exchanging signals. For static transmission without additional frequency conversion, the local frequencies at emission position e and reception position r satisfy

$$\frac{\nu_r}{\nu_e} = \frac{A(x_e)}{A(x_r)}. \tag{2.12}$$

When a signal travels from a region of lower clock rate to a distant receiver, the receiver reads a lower frequency. Its leading form connects to the weak-field gravitational-redshift relation [7]. Force and redshift thus share the same potential: the former compares phase gradients between neighboring positions, while the latter compares phase accumulation rates at different positions.

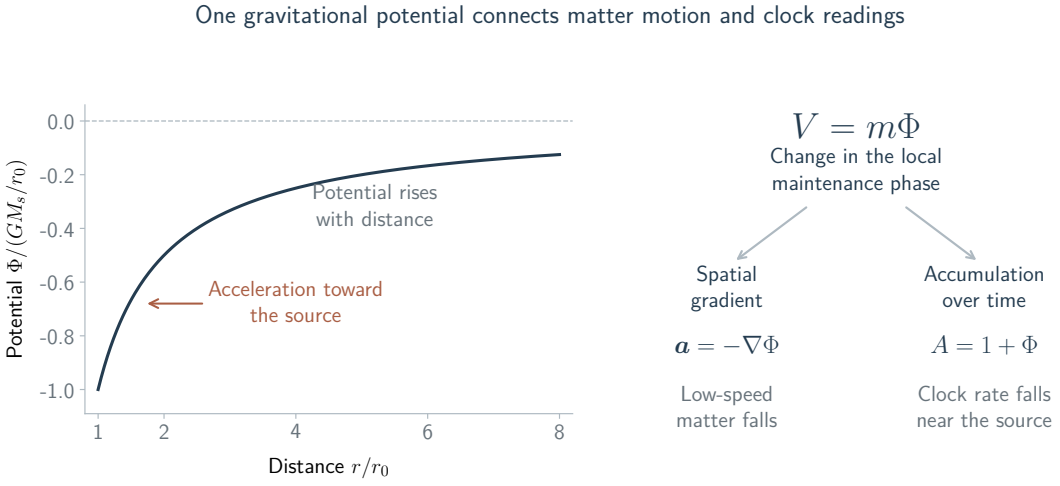


Figure 2.1. Gravitational potential, motion, and clock rates. The exterior potential increases with distance, and its negative gradient accelerates slow matter toward the source. The same negative potential lowers the local maintenance frequency, so clocks accumulate less phase than the distant reference. The curve is a normalized illustration of the weak-field far region; its vertical axis does not represent the full nonlinear result near the source.

§2.5 Spatial Length Scales and Geometric Representation

Clocks and rods read how the same background affects different actions of matter. Internal maintenance phase determines the local clock rate, whereas spatial exchange determines the propagation principal part. Both must be computed separately from one complete reception process.

Let B be the work package, V_x the background package at position x, and S_x the swap of these complete packages. Whether records travel with a package is part of the actual deployment. Use the position-controlled swap and return $\mathcal{C} = \sum_x P_x \otimes S_x$, with participating-response projector R_B. Complete reception is $H_C = \mathcal{C}(K \otimes R_B)\mathcal{C}$. For independent input package states $\sigma_B, \sigma_x, \sigma_y$, the local and edge responses are

$$\boxed{a_x = \mathrm{Tr}(R\sigma_x), \qquad t_{yx} = \mathrm{Tr}(R\sigma_x\sigma_B\sigma_y).} \tag{2.13}$$

These contractions follow from complete package exchange. The ordered three-package product cannot be replaced by a single occupation. With the original matter mass action $M_f Y$ and motion matrices α_i, the coefficients of the conditional mean maintenance and spatial actions are determined by a_x and t_{yx}.

An explicit preparation uses a two-state occupation unit as each complete exchanged package, with independent inputs, a common available state $|0\rangle$, $R = |0\rangle\langle 0|$, and $\sigma_i = \mathrm{diag}(h_i, 1 - h_i)$. Archived records remain outside the exchanged packages. Then $a_x = h_x$ and $t_{yx} = h_B h_x h_y$. If records or other internal degrees of freedom travel with a package, the complete trace must be evaluated again; diagonal occupation alone does not imply this product. The corresponding self-adjoint action for the stated preparation is

$$\overline{H} = M_f Y\, h(X) + \nu h_B \sum_i \alpha_i h(X) D_i h(X). \tag{2.14}$$

The order hD_ih retains gradient contributions in a nonuniform background. With M actual independently prepared work arrays, simultaneously coupled with strength $1/M$ each, the difference between this mean action and the complete reduced output can be controlled over a finite time window. Arrays, records, and preparation have real resource costs; the environment is not reset during evolution. Section D6 gives the error and phase-readout bounds.

In a locally uniform, long-wavelength window of nonzero availability, calibration against the same-source field-free background h_0 gives

$$A = \frac{h}{h_0}, \qquad C_m = \frac{h^2}{h_0^2}, \qquad E_+(x,p) \simeq \sqrt{m_0^2 A^2 + c_0^2 C_m^2 |p|^2}. \tag{2.15}$$

Use the reference units $c_0 = 1$ of this chapter. If the spatial action of light is received through the same type of package, with a common field-free speed calibration, then these two quantities admit the geometric representation [8]

$$ds^2 = A(x)^2 dt^2 - B(x)^2 d\boldsymbol{x}^2. \tag{2.16}$$

$$E(x,p) = A(x)\sqrt{m_0^2 + \frac{|p|^2}{B(x)^2}}, \qquad \left|\frac{d\boldsymbol{x}}{dt}\right|_{m_0=0} = \frac{A}{B}. \tag{2.17}$$

Comparing the actual principal part and mass shell term by term gives the general relation and its value for this preparation:

$$\boxed{B = \frac{A}{C_m}, \qquad C_m = A^2 \implies B = A^{-1}.} \tag{2.18}$$

The reciprocal relation follows jointly from single-package maintenance and three-package overlap. Other preparations give different outputs: the original coherent phase package and the first-order common acceptance response can give $A = C_m$, whereas gating maintenance without changing displacement gives $C_m = 1$. Each configuration has its own actual action and cannot substitute for another. A common activity fraction cancels in relative coefficients only when maintenance and spatial action use the same clock.

For static nonuniform light propagation $hH_{\gamma 0}h$, the preserved transverse condition is weighted: $\mathrm{Div}(h^{-1}\psi_\gamma) = 0$. Emission must enter this propagating space. A time-dependent h introduces further constraint terms. The conditional realization of Eq. (2.18) therefore does not automatically establish common geometry in a persistent time-dependent or strong background. The companion Geometric Extension and Optical Examples computes consequences of specified clock–rod and nonlinear constitutive relations; its exponential profile still requires an independent transport basis.

§2.6 Nonlinear Distributions

In a weak field, background exchange can be calculated with a common reference. Near a source, background changes may also affect exchange rates and stationary weights at different positions, so transport itself varies with the field. A nonlinear field equation must incorporate this dependence into the existing flow relation.

Let n_i be the transported background deficit and $\pi_i > 0$ the stationary weight of the same network. For exchange satisfying detailed balance, let w_{ij} be the jump rate from i to j, with conductance $\kappa_{ij} = w_{ij}\pi_i = w_{ji}\pi_j$. The net flow from i to j and the stationary source equation are [9]

$$J_{ij} = \kappa_{ij}\left(\frac{n_i}{\pi_i} - \frac{n_j}{\pi_j}\right), \qquad \sum_j J_{ij} = s_i. \tag{2.19}$$

This form retains local occupation, stationary weight, and exchange capacity separately. Constant coefficients recover the harmonic distribution above. When the coefficients vary with the field, flow is conserved in the same way, but the spatial profile changes accordingly.

The nonlinear profile depends on how n_i, π_i, and κ_{ij} vary jointly with the background state. Given these relations, the sources, and the boundary conditions, Eq. (2.19) determines the net flow and stationary distribution. Section S1 of the supplement gives a constitutive example yielding an exponential distribution. Dynamical Supplement D7 retains the occupation covariance and exchange coherence produced by persistent sources and transport, and bounds deviations from a time-ordered mean action. Without control of those correlations, one occupation variable does not close clock–rod and spatial propagation responses.

§2.7 Time-Dependent Gravitational Fields

After a source changes, new writing enters the background cycle by cycle and propagates together with the existing distribution. A static field is the stationary result of this process under a fixed source, while the time-dependent response retains the full build-up history. For the refreshing realization of Chapter 1, write $T = R(I - D_\varepsilon)$ and $\boldsymbol{q}_n = RD_\varepsilon \boldsymbol{b}_n$. Then

$$\boldsymbol{\xi}_N = T^N \boldsymbol{\xi}_0 + \sum_{n=0}^{N-1} T^{N-1-n} \boldsymbol{q}_n. \tag{2.20}$$

Each term represents the share of one source-writing contribution that arrives after subsequent exchanges. When the source stops writing a bias but unbiased refreshing continues, the existing bias keeps evolving under T and relaxes toward the supply state. When the source varies periodically, outputs with different delays combine to form the response. The number of propagation cycles and the total elapsed duration of each cycle jointly determine arrival and decay.

A static $1/r$ profile alone does not determine time-dependent behavior. For the present occupation process, $R = I - cL$ outside the source. Converting cycles to time with the complete cycle period τ gives, in the slowly varying limit,

$$\partial_t \xi = D\nabla^2 \xi, \qquad D = \frac{c\ell_x^2}{\tau}. \tag{2.21}$$

This describes transport and relaxation of the mean occupation. It can establish a harmonic static far field, while its time dependence is diffusive. The finite sequence of exchanges still specifies which positions changes reach step by step; the continuum diffusion equation describes their long-time, large-scale average [10].

Coherent propagation additionally retains amplitudes and relative phases. The original field model [11] connects gravitational radiation to directional perturbations of the common time and length scales, read through the propagation responses of light and matter. Section 5.4 and Dynamical Supplement D8 give a separate finite propagation preserving directional constraints; its material connection to the original common clocks and rods remains to be established. Equation (2.21) describes relaxation of the mean occupation.

Energy transfer in radiation remains part of the joint accounting. Changes in the source, propagating background, receiving matter, and preparation apparatus together determine the output. Where controlled actions are used, preparation and switching work must also be included.

The local maintenance phase connects gravitational potential, motion, and clock rates. Coherent exchange between charged matter systems additionally involves internal states and their phase changes along paths.

§3 Gauge Fields and Nuclear Interactions

Electromagnetic, weak, and color interactions select exchange channels through the charges of matter. A source injects excitations and phase changes into the background, whose propagation carries them to compatible matter. Phase comparisons between positions, internal-state transfers, and momentum exchange together constitute the physical content of these fields. Interactions between nucleons likewise proceed through emission, propagation, and absorption, with internally structured composite matter at the endpoints.

§3.1 Charges and Exchange

Charge specifies how matter participates in a type of interaction. For electromagnetism, take the self-adjoint operator Q in the complete matter representation as the electric-charge generator. Charge states satisfy $Q|a\rangle = q_a|a\rangle$, where q_a includes the elementary charge unit. Matter at both ends and the background transmitting the excitation participate jointly in exchange, and their charge changes must balance in the same process. An actual charge-preserving joint action satisfies

$$[\mathbb{U}_{\mathrm{ex}}, Q_A + Q_B + Q_{\mathcal{B}}] = 0, \qquad q'_A + q'_B + q'_{\mathcal{B}} = q_A + q_B + q_{\mathcal{B}}. \tag{3.1}$$

The second relation applies to nonzero transfer matrix elements between states of definite charge. Photons carry no electric charge: electromagnetic exchange can transfer energy and momentum while preserving the charges at both ends. Transfers involving charged modes allow the endpoint charges to change, with the exchanged component balancing the difference. Both cases are described by the same joint balance.

References [2] and [3] first establish multiplicative conservation of character charge in the joint state. The physical counterparts of electric, weak, and color charges follow Ref. [3]; this chapter uses these generators in the corresponding matter representations. Character labels specify types and transfer directions, while charge values are determined by Q. The opposite charges of antiparticles are likewise determined in the complete representation. Type conjugation, exchange direction, and charge sign thus retain their respective meanings.

Weak and color actions also change the internal states of matter. The two- and three-dimensional factors of Ref. [3] provide the structure for these matrix actions. Writing $E_{ab} = |a\rangle\langle b|$ for a transfer within a factor gives

$$[E_{ab}, E_{cd}] = \delta_{bc}E_{ad} - \delta_{da}E_{cb}. \tag{3.2}$$

This relation explains why two successive transfers can depend on their order. Off-diagonal transfers together with relative diagonal phases provide three traceless self-adjoint directions for a two-dimensional factor and eight for a three-dimensional factor. These carry the matrix structures of weak and color interactions. Electromagnetic action is organized by the charge generator retained in the neutral action, and its actual response is read through relative phases and the motion of charged matter.

Source strength comes from the actual matrix elements and elapsed durations of these actions. An emission may change the internal source state, or it may leave the

source unchanged after mediator retrieval while changing correlations between the two ends. Reference [3] supplies a joint realization in which the mediator regains its prepared state while matter still undergoes transfer. A persistent field connects such exchange to multiple matter systems and continuing propagation: the source supplies excitations and phases in the relevant channels, receiving matter responds through corresponding vertices, and the source itself undergoes back-reaction. The coupling coefficients calibrate the relative strengths of these vertices; propagation additionally determines the spatial distribution.

§3.2 Phase and Propagation

As matter passes through the background, its internal state changes segment by segment through successive exchanges. For the part of propagation that retains coherence, a transfer matrix T_{ji} from position i to j can represent one action. Along a history γ, the segment matrices multiply in their actual order to give T_γ. In a one-dimensional internal representation, this action can be a phase; in a multidimensional representation, it can also rotate the internal state of matter.

Internal states at different positions must be expressed in their respective bases. If the components at position i are written as $\psi'_i = G_i\psi_i$, with G_i unitary, the same actual transfer is represented by

$$T'_{ji} = G_j T_{ji} G_i^\dagger, \qquad T'_\gamma = G_{\text{end}} T_\gamma G_{\text{start}}^\dagger. \tag{3.3}$$

Basis changes at neighboring segments cancel inside the product, leaving only transformations at its endpoints. This is the meaning of a local gauge representation in propagation: internal coordinates may be chosen at each position, and the actions connecting positions must change representation accordingly. When matter, background action, and readout are transformed together, event probabilities remain invariant.

For two coherent histories with the same start and end, their relative transfer is $T_{\gamma_2}^\dagger T_{\gamma_1}$. It transforms by conjugation at the start, so the interference term evaluated with the initial state is invariant. For a selected detection outcome, if the histories contribute amplitudes a_1, a_2 and leave normalized environment states $|e_1\rangle, |e_2\rangle$, its probability contains

$$p = |a_1|^2 + |a_2|^2 + 2\,\mathrm{Re}[a_1\overline{a_2}\langle e_2|e_1\rangle]\,. \tag{3.4}$$

The relative phase is produced by the two actual evolutions, while visibility also depends on whether the environment distinguishes them [2, 12]. When coherent retrieval restores a common environment preparation, the path phases can continue to interfere in matter. When records distinguish the histories, the cross term changes with their environment overlap. Field phase and the final record therefore belong to the same process.

When coherent propagation is smooth on the observation scale and local actions admit a unitary-connection description, set $\hbar = c_0 = 1$ and write infinitesimal transfer as [13]

$$D_\mu = \partial_\mu + i\mathcal{A}_\mu, \qquad T_\gamma = \mathcal{P}\exp\left(-i\int_\gamma \mathcal{A}_\mu dx^\mu\right). \tag{3.5}$$

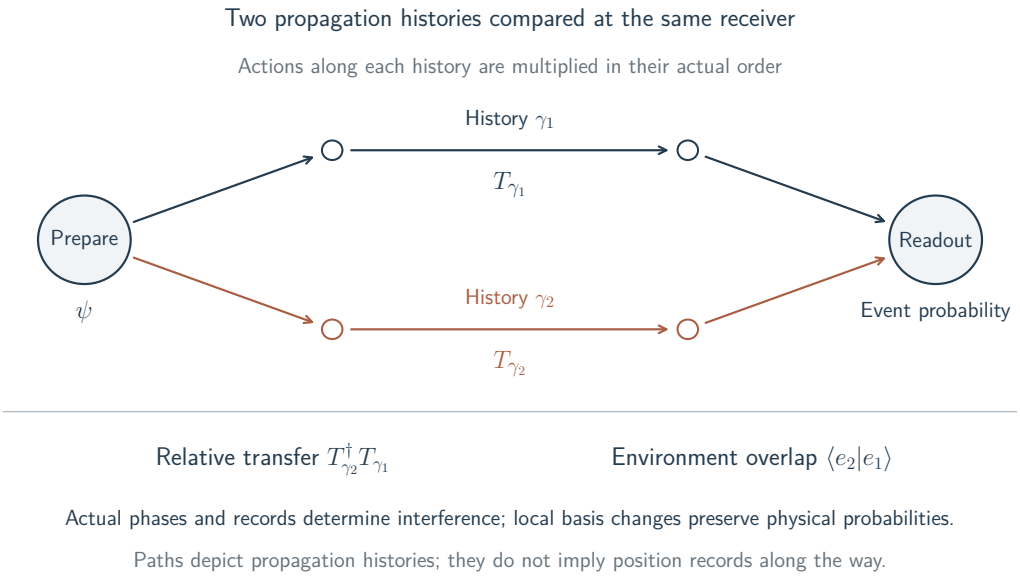


Figure 3.1. Comparing propagation phases. Two histories connect the same preparation and receiver, each accumulating its actual transfer matrix; together they affect readout at the receiver. A local basis change transforms the representations of adjacent transfer matrices without changing physical probabilities. The paths depict two propagation histories of amplitudes, without implying pointwise measurements along the way. Whether the environment distinguishes them is determined by actual records.

Here $\mathcal{A}_\mu$ is a self-adjoint matrix containing the coupling and charge generators, and $\mathcal{P}$ orders actions along propagation. The derivative D_μ compares matter components using actual transfer between neighboring positions. To preserve Eq. (3.3), the connection transforms as $\mathcal{A}'_\mu = G\mathcal{A}_\mu G^\dagger + i(\partial_\mu G)G^\dagger$. The phase connection thereby compares local representations; amplitude changes and dissipation during propagation remain calculable from the full exchange.

Comparing two orders of transfer around a small loop gives a leading difference determined by the commutator of covariant derivatives:

$$[D_\mu, D_\nu] = i\mathcal{F}_{\mu\nu}, \qquad \mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu + i[\mathcal{A}_\mu, \mathcal{A}_\nu]. \tag{3.6}$$

The curvature $\mathcal{F}_{\mu\nu}$ of this propagation transforms as $\mathcal{F}'_{\mu\nu} = G\mathcal{F}_{\mu\nu}G^\dagger$. In an Abelian action, the matrices commute and loop differences arise from spatial and temporal variations of the connection. Internal weak and color actions additionally retain the commutator arising from their ordering. Phase changes and internal transfers thus acquire a common field description while preserving their distinct action structures.

Connection and curvature describe the geometry of coherent transfer. Their evolution is determined by actual background action. We next construct finite flux return, complete massive transport, and color-link return, then read continuum field equations in their respective controlled windows. Mass and charge identifications are inherited from Ref. [3]. Auxiliary preparation, spatial coupling, and scale separation remain explicit conditions; standard continuum forms serve to compare the derived responses.

§3.3 Electromagnetic Fields

Electromagnetic interaction connects charge transfer in matter to coherent changes in the background. As charge moves along an edge, the background carries the corresponding flux; a completed matter loop can also leave a divergence-free background change. Static storage and energy-carrying propagation must be determined together from the actual evolution of these quantities. We construct this response through finite joint exchanges; the Maxwell form follows in a controlled continuum representation.

The field equations below first use lattice units and band slow time τ; actual time and lattice spacing are restored together at the end. Fix a fundamental flux-charge unit $q > 0$, and first consider the branch of charge q. Take $\mathcal{A}_\mu = qA_\mu$, with $A_\mu = (\phi, -\boldsymbol{A})$ and signature $(+,-,-,-)$. The coherent transfer phase and curvature are [14]

$$\varphi_\gamma = q \int_\gamma (\boldsymbol{A} \cdot d\boldsymbol{x} - \phi \, d\tau), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{3.7}$$

The temporal–spatial and spatial–spatial differences of the same connection give

$$\boldsymbol{E} = -\nabla\phi - \partial_\tau \boldsymbol{A}, \qquad \boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{3.8}$$

To determine its dynamics, let D and C be the vertex–edge and edge–face incidence maps of the actual spatial graph, with $DC^\dagger = 0$. Each edge retains a finite integer flux N_e and its truncated raising shift U_e; W_p is the ordered product of raising and lowering operators around a face. Charged matter transfer simultaneously lowers the corresponding edge flux, so the complete transfer preserves

$$G_x = (DN)_x - \rho_x/q, \qquad [G_x, H_m] = 0, \qquad \partial_\tau \rho = -Dj. \tag{3.9}$$

Here H_m includes the original matter maintenance and joint charged hopping, and j is the current determined by that same hopping. Failed outputs at the flux cutoff remain in the complete space. An input satisfying $G_x = 0$ must subsequently evolve under a constraint-preserving action.

Background storage follows from two actual returns. Edge flux controls a neutral two-state auxiliary; loop transport and a reference arm enter the same output of another auxiliary with opposite phase. With $Y_r|\pm\rangle = \pm|\pm\rangle$ and $\sigma_-|+\rangle = |-\rangle$, take

$$\begin{aligned} H(\delta) &= H_R + \delta V + \delta^2 H_m, \qquad H_R = \sum_r m_r Y_r, \\ V &= \sum_e \lambda_e X_e \otimes N_e \\ &\quad + \sum_p \lambda_p \big[\sigma_{-,p} \otimes (I - W_p) + \sigma_{+,p} \otimes (I - W_p)^\dagger\big]. \end{aligned} \tag{3.10}$$

The positive m_r and the parameter δ specify a separation between fast maintenance and slow response. The finite exchanges of the mathematical foundation and the preparation and protection of Ref. [3] provide tools for implementing these matrix elements. Deployment on each edge and face, and preparation of the auxiliaries, are physical conditions of this construction. All auxiliaries and returns participate

in the evolution. For fixed parameters, $[H, G_x] = 0$, and the energy of the complete H is conserved.

Prepare the isolated spectral band continuously connected to the all-upper input. Leaving that input and returning through the same auxiliary gives the positive denominator $2m_r$; second-order cross returns through distinct auxiliaries vanish. The band action is therefore

$$\boxed{\begin{aligned} H_{\mathrm{band}} &= E_R + \delta^2 H_2 + O(\delta^4), \\ H_2 &= H_m + \frac{1}{2}\sum_e a_e N_e^2 + \frac{1}{2}\sum_p b_p (I - W_p)^\dagger (I - W_p), \\ a_e &= \lambda_e^2/m_e, \qquad b_p = \lambda_p^2/m_p. \end{aligned}} \tag{3.11}$$

The positive sign follows from this spectral preparation. The upper band has an energy and preparation cost; it is not the thermal ground state of the complete system. Choosing the lower branch changes the sign of the return shift. Section D2 of the Supplementary Dynamical Proofs gives the band isometry, remainder, and cutoff terms. Energies here refer to the corresponding actual generators; their relation to bare-shell energy requires a separate accounting of the complete implementation.

For uniform $a_e = a$ and $b_p = b$, write $S_p = (W_p - W_p^\dagger)/(2i)$, $K_p = (W_p + W_p^\dagger)/2$, and use slow time $\tau = \delta^2 t$. Define

$$E_e = aN_e/q, \qquad B_p = -S_p/q, \qquad \varepsilon = q^2/a, \qquad \mu^{-1} = q^2 b. \tag{3.12}$$

Commutators with the same H_2 give the finite equations

$$DE = \rho/\varepsilon, \qquad \varepsilon \dot{E} = \mu^{-1} C^\dagger B - j, \qquad \dot{B} = -CE + \mathcal{R}_B, \tag{3.13}$$

where

$$\begin{aligned} \mathcal{R}_{B,p} &= -\frac{a}{2q}\sum_e C_{pe}\{N_e, K_p - I\} - \frac{i}{q}[H_m + H_B, S_p], \\ H_B &= \frac{b}{2}\sum_p (I - W_p)^\dagger (I - W_p). \end{aligned} \tag{3.14}$$

Small loop phases, controlled flux fluctuations, and controlled cutoff boundaries make $\mathcal{R}_B$ negligible relative to the leading terms. Then $W_p \simeq e^{i(C\theta)_p}$ and $A_e = -\theta_e/q$ connect the finite readouts to Eq. (3.8). In a uniform three-dimensional long-wavelength window, this gives the Maxwell form [15]:

$$\boxed{\begin{aligned} \nabla \cdot \boldsymbol{E} &= \rho/\varepsilon, \quad \mu^{-1}\nabla \times \boldsymbol{B} - \varepsilon \partial_\tau \boldsymbol{E} = \boldsymbol{j}, \\ \nabla \cdot \boldsymbol{B} &= 0, \qquad\qquad \nabla \times \boldsymbol{E} + \partial_\tau \boldsymbol{B} = 0. \end{aligned}} \tag{3.15}$$

The absence of magnetic sources follows here from cancellation of loop boundaries. At finite phases, the linear divergence of S_p need not vanish exactly. These approximations must remain valid over the time interval considered; a small initial phase alone does not guarantee this.

The same electric flux energy determines static attraction and repulsion. On a connected graph of zero total charge, let $L = DD^\dagger$ and $N_L = D^\dagger L^+ \rho/q$. Orthogonal decomposition gives

$$H_E = \frac{1}{2\varepsilon}\rho^\dagger L^+ \rho + \frac{a}{2}\|N - N_L\|^2. \tag{3.16}$$

In the noncompact approximation with localized phases, flux translation preserves the loop action and shifts the entire transverse spectrum by this longitudinal energy. Finite integer flux retains sector and boundary corrections. In a three-dimensional far-field continuum window, the point-source kernel of L^+ is $1/(4\pi r)$, giving

$$\phi(r) = \frac{q_s}{4\pi\varepsilon r}, \qquad V(r) = \frac{q_s q_t}{4\pi\varepsilon r}, \qquad \boldsymbol{F} = \frac{q_s q_t}{4\pi\varepsilon r^2}\widehat{\boldsymbol{r}}. \tag{3.17}$$

Like charges repel and opposite charges attract through the gradient of the actual energy. The same phase in charged hopping gives $H_m(A) = H_m(0) - \sum_e A_e j_e(0) + O(A^2)$. For a single-band wave packet in a slowly varying field, hold the other material parameters fixed and apply Hamilton's equations to its actual semiclassical symbol $\epsilon_m(\boldsymbol{p} - q\boldsymbol{A}) + q\phi$. This gives

$$\boxed{\frac{d\boldsymbol{\pi}}{d\tau} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}), \qquad \boldsymbol{\pi} = \boldsymbol{p} - q\boldsymbol{A}.} \tag{3.18}$$

The source current and force are thus determined by one matter–background action. If the intrinsic band has additional explicit position dependence, the force also contains $-(\nabla_x \epsilon_m)_{\boldsymbol{\pi}}$. Interband transitions, finite phases, and lattice gradients remain corrections to this semiclassical expression.

Energy exchanged between electric edges and magnetic faces cancels term by term, while work on matter has the opposite sign. In the same continuum window, the field energy flux is [16]

$$\boxed{\partial_\tau \left(\frac{\varepsilon}{2}|\boldsymbol{E}|^2 + \frac{1}{2\mu}|\boldsymbol{B}|^2 \right) + \nabla \cdot \left(\frac{\boldsymbol{E} \times \boldsymbol{B}}{\mu} \right) = -\boldsymbol{j} \cdot \boldsymbol{E}.} \tag{3.19}$$

Quantum readouts use symmetrized products; products of means do not replace general correlated expectations. The complete model retains auxiliary and contact energies. The quadratic field energy is the leading band expression at weak phase.

In the same fundamental charge unit, an actual edge transfer of integer charge $Q = nq$ must carry flux action $L_{n,e} = (U_e^\dagger)^n$ for $n \geq 0$, or $L_{n,e} = U_e^{-n}$ for $n < 0$, where the latter exponent is a positive integer. Thus $[N_e, L_{n,e}] = -nL_{n,e}$. The joint jump $T_{n,e} = h_{n,e} \otimes L_{n,e}$ preserves the same constraint $DN - \rho/q$, with current $j_{n,e} = iQ(T_{n,e}^\dagger - T_{n,e})$. Shifts beyond a finite flux boundary vanish. Extending the charge branch requires this actual deployment.

Propagation and coupling calibration constrain the same coefficients. Let ℓ be the cubic lattice spacing and s the common conversion rate from full reference time to band time; in the autonomous Eq. (3.10), $s = \delta^2$. Transverse waves and the static energy difference in the same charge channel give

$$\boxed{\begin{aligned} &\omega_\tau^2(k) = 4ab \sum_i \sin^2(k_i \ell/2), \\ &c_{\text{field}} = s\ell\sqrt{ab}, \qquad \alpha_n^{\text{field}} = \frac{n^2}{4\pi}\sqrt{a/b}. \end{aligned}} \tag{3.20}$$

The corresponding physical readouts are $A_{\mathrm{phys}} = A/\ell$, $\phi_{\mathrm{phys}} = s\phi$, $E_{\mathrm{phys}} = sE/\ell$, and $B_{\mathrm{phys}} = B/\ell^2$, with source readouts $\rho_{\mathrm{phys}} = \rho/\ell^3$ and $j_{\mathrm{phys}} = sj/\ell^2$. The coefficients transform together as $\varepsilon_{\mathrm{phys}} = \varepsilon/(s\ell)$ and $\mu_{\mathrm{phys}} = \mu/(s\ell)$. In subsequent actual-space-and-time formulas, ε, μ denote these physical coefficients. Here n is an integer in the same fundamental charge unit, and $\hbar = 1$. Matching the light speed and corresponding coupling level of Ref. [3] requires both a product relation and a ratio relation. A common time conversion or a common matching factor on both returns leaves the dimensionless ratio unchanged. How the original matching process fixes a/b remains undetermined. The construction therefore establishes conditional electromagnetic dynamics and a calibration criterion, without treating the unresolved relative strength as a derived constant.

§3.4 Weak-Interaction Fields

Weak interaction changes the internal state of matter during transmission. Denote the three traceless self-adjoint generators of the two-dimensional factor by T_a; the charges of $T_\pm = T_1 \pm iT_2$ are determined by the complete charge generator. Internal transfers at the source and receiver balance the charge carried by the mediator. Neutral directions retain the coherent mass construction of Ref. [3], represented in the electroweak neutral mixing basis [17, 18]:

$$|\gamma\rangle = s_v|w\rangle + c_v|y\rangle, \qquad |Z\rangle = c_v|w\rangle - s_v|y\rangle, \quad s_v^2 + c_v^2 = 1. \tag{3.21}$$

The two basis directions are orthonormal, with a common kinetic normalization. The relation $s_v^2 = \alpha_{\mathrm{EM}}^{(0)}/\alpha_W^{(0)}$ uses the original Born-level vertex-response ratio. The common mass output couples only to the Z direction, so

$$\mathsf{M}^2_{\mathrm{neutral}} = M_Z^2|Z\rangle\langle Z|, \qquad \mathsf{M}^2_{\mathrm{neutral}}|\gamma\rangle = 0. \tag{3.22}$$

This mass identification does not by itself prescribe spatial transport. Let $\mathcal{M}$ be the self-adjoint bidirectional action of the original complete mass exchange, and Ξ its input–output orientation, with $\{\mathcal{M}, \Xi\} = 0$. On a three-dimensional periodic lattice, set $d_i = S_i - I$. Three vector inputs are transported through one divergence and three curl auxiliaries:

$$Q = \begin{pmatrix} d^\dagger \\ [d]_\times \end{pmatrix}, \qquad D_Q = \begin{pmatrix} 0 & Q^\dagger \\ Q & 0 \end{pmatrix}, \qquad Q^\dagger Q = LI_3, \quad L = \sum_i d_i^\dagger d_i. \tag{3.23}$$

Each auxiliary retains the same mass carrier, and transport is controlled by Ξ. The actual finite action and its square are

$$\boxed{h = \mathcal{M} \otimes I_7 + \nu\Xi \otimes D_Q, \qquad h^2 = \mathcal{M}^2 \otimes I_7 + \nu^2 I \otimes D_Q^2.} \tag{3.24}$$

The three vector inputs on a mass-bright branch $M_j > 0$ therefore share the dispersion $\Omega_j^2 = M_j^2 + \nu^2 L(k)$. All frequencies use the same actual time unit; masses already converted to complete-cycle units in Ref. [3] are not rescaled again. Spatial auxiliaries carry transport and are not counted as additional physical polarizations. Original mass-dark branches remain, with participation determined by the actual source.

Let J_j select the complete positive rest branch at the source and receiver. Equation (3.24), together with the zero input diagonal block of D_Q, gives

$$\boxed{R_j(z) = J_j^\dagger (z-h)^{-1} J_j = \frac{z + M_j}{z^2 - M_j^2 - \nu^2 L} I_3.} \tag{3.25}$$

The numerator follows from the input–output coherence retained at the actual endpoint. Coupling only the bare mass input instead gives numerator z, changing the zero-frequency response. Source, receiver, and return form one joint generator with endpoint response $\Sigma(z) = V^\dagger (z-h)^{-1} V$. Both self returns and cross exchange must remain; the radial denominator alone is insufficient.

For $M_j/\nu \ll 1$ and long-wavelength sources and receivers far from boundaries, set $c = \nu\ell$. The static denominator approaches the modified Helmholtz operator, whose continuum point-source kernel is [19]

$$(-c^2\nabla^2 + M_j^2)\mathscr{G}_j = \delta^{(3)}(\boldsymbol{x}), \qquad \boxed{\mathscr{G}_j(r) = \frac{e^{-M_j r/c}}{4\pi c^2 r}.} \tag{3.26}$$

The actual static endpoint kernel is $-M_j\mathscr{G}_j$, combined with both endpoint strengths, orientations, and source shapes. Section D3 gives the volume normalization from lattice to continuum point sources. Mass determines the decay length c/M_j; the endpoints determine transfer type and strength. Finite lattices, transport cutoffs, and high-frequency inputs require Eq. (3.25). This construction establishes a conditional mechanism for massive propagation and short-range exchange; it does not determine weak chirality, Fermi normalization, or decay widths.

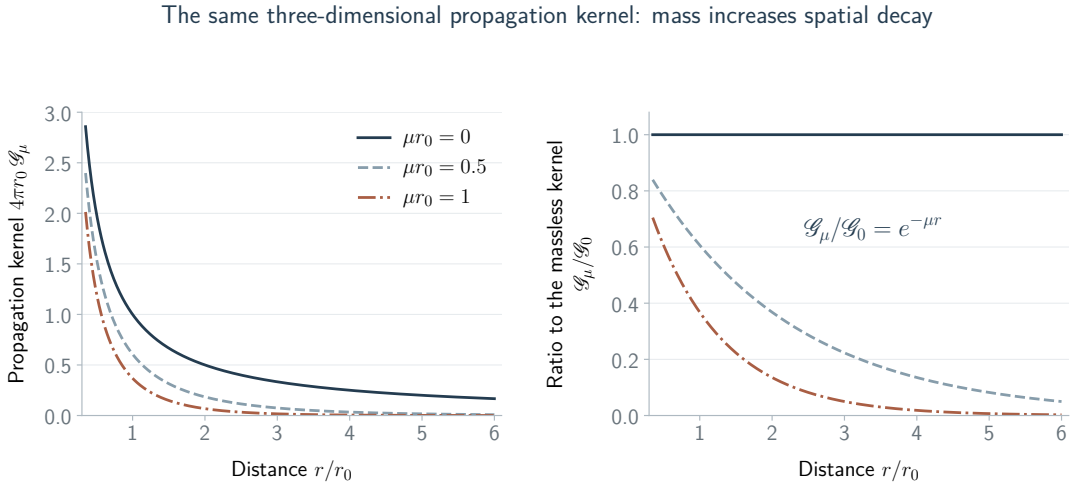


Figure 3.2. Mass and spatial interaction range. With propagation speed set to one, the figure compares the radial kernel in Eq. (3.26) and its exponential factor relative to the massless kernel. Parameters are dimensionless illustrations. The actual endpoint numerator, charge transfer, and spin response remain determined by the same joint process.

§3.5 Color Fields

The order of internal color actions enters the propagation amplitude. Retain the three-dimensional color representation of Ref. [3], with self-adjoint traceless generators t^a normalized by $\mathrm{tr}(t^a t^b) = \delta^{ab}/2$. A dynamical color connection must itself be a background object that matter can change.

Assign a finite link register to each directed edge. The operator matrix U_e transfers endpoint color, while L_e^a, R_e^a record the endpoint actions:

$$[L^a, U_{ij}] = -t^a_{ik} U_{kj}, \qquad [R^a, U_{ij}] = U_{ik} t^a_{kj}, \qquad [L^a, R^b] = 0. \tag{3.27}$$

Section D4 gives an explicit finite realization and its representation cutoff. Matter hopping is $\psi_x^\dagger U_e \psi_y + \mathrm{h.c.}$, so matter and link charges jointly satisfy Gauss constraints. The nonunitarity defect of finite U_e remains in the complete action.

Form the ordered loop W_p on each actual face. Electric return couples L_e^a to an auxiliary carrying the corresponding adjoint charge. Magnetic return couples all nine orthogonal components of $I - W_p$ to singlet and adjoint auxiliaries. Their color charges enter the complete Gauss generator. The same isolated upper-band return used for electromagnetism gives

$$\boxed{\begin{gathered} H_{2,c} = H_{m,c} + \frac{\alpha}{2} \sum_{e,a} (L_e^a)^2 + \frac{\beta}{2} \sum_p \mathrm{tr}[(I - W_p)^\dagger (I - W_p)], \\ \alpha = \frac{\lambda_E^2}{m_E}, \\ \beta = \frac{\lambda_B^2}{m_B}. \end{gathered}} \tag{3.28}$$

The return energy is the sum of the Gram operators of the actual outputs. Keeping the full matrix preserves noncommuting information; taking the trace of $I - W_p$ before squaring loses the leading traceless curvature. The complete action conserves total color charge and energy. Electric flux and loop energies can exchange even without an externally prescribed source.

In a smooth, nearly unitary window with controlled cutoff, write $U_i \simeq e^{i\ell A_i}$. Expansion in the specified path order gives

$$W_{ij} = I + i\ell^2 F_{ij} + O(\ell^3), \qquad F_{ij} = \partial_i A_j - \partial_j A_i + i[A_i, A_j]. \tag{3.29}$$

Substitution in Eq. (3.28) gives leading magnetic Gram term $\ell^4 \,\mathrm{tr}\, F_{ij}^2$. With continuum conjugate variable $\Pi_i^a = -L_e^a/\ell^2$, counting each unoriented face once gives $\kappa_E = \alpha\ell$ and $\kappa_B = \beta\ell/2$. The same band action yields

$$\boxed{\begin{gathered} \partial_\tau A_i = \kappa_E \Pi_i, \\ \partial_\tau \Pi_i = \kappa_B D_j F_{ji} - j_i, \\ D_i \Pi_i = \rho, \\ D_i X = \partial_i X + i[A_i, X]. \end{gathered}} \tag{3.30}$$

This is the Yang–Mills principal part of the constructed response [20]. Source current comes from the same matter hopping, and covariant charge balance is com-

patible with the field equations. The commutator expresses the field's own participation in color transfer. The finite model uses complete ordered loops and actual commutators; continuum notation does not remove its cutoff corrections.

Color singlets, gluon directions, and confinement in the specified separation process remain inherited from Ref. [3]. This section supplies finite color-field dynamics and its continuum working window. It does not identify the minimal finite encoding with a continuum limit or add claims of universal confinement, a nonperturbative mass gap, or coupling running.

§3.6 Nuclear Interactions

Nuclear interaction connects two complete nucleons. The pion mass determines the propagation scale, whereas nucleon spin, flavor, and spatial coupling determine direction and attraction or repulsion. Both endpoints remain present throughout, and the nucleons and pion jointly carry the charge changes in emission and absorption.

Let J_N embed the four-state nucleon carrier of Ref. [3]. On its three constituent slots, define $\mathsf{S}_i = \sum_j \sigma_j^i$ and $\mathsf{T}_a = \sum_j \tau_j^a$. The complete nucleon satisfies

$$\mathsf{S}_i \mathsf{T}_a J_N = J_N(\tau_a \otimes \sigma_i). \tag{3.31}$$

This collective action preserves the four nucleon states and differs from a sum of spin–flavor products on individual slots. The pion retains its original color singlet, pseudoscalar 0^- identification, and three-flavor carrier. Equal, oppositely phased spatial arms preserve parity and enter the same complete output, giving the actual momentum vertex

$$V_a(k) = g_\pi \tau_a \, \boldsymbol{\sigma} \cdot \widehat{\boldsymbol{k}}, \qquad \widehat{k}_i = \sin(k_i \ell)/\ell. \tag{3.32}$$

Spin and flavor arise from the complete nucleon action, while the odd momentum factor comes from cancellation between neighboring arms. The coefficient g_π is the actual exchange strength. Equal strengths on three axes and flavor covariance are deployment conditions, not consequences of charge conservation alone.

The pion uses its original complete mass exchange with one scalar spatial input and three gradient transport auxiliaries. Its complete positive rest input has propagation matrix element

$$R_\pi(z) = \frac{z + m_\pi}{z^2 - m_\pi^2 - \nu^2 L}, \qquad R_\pi(0) = -m_\pi (m_\pi^2 + \nu^2 L)^{-1}. \tag{3.33}$$

Two nucleons and a zero/one-pion capacity form the declared finite system. With $V = V_A + V_B$, its joint generator and complete return are

$$\boxed{H_{\text{joint}} = \begin{pmatrix} H_{NN} & V^\dagger \\ V & H_Q \end{pmatrix}, \qquad \Sigma(z) = V^\dagger (z - H_Q)^{-1} V.} \tag{3.34}$$

The block H_Q retains both nucleons and the pion's mass, position, and transport auxiliaries. Self returns and cross returns coexist. Exchange paths not distinguished by the complete record add as amplitudes. Joint charge and total energy are conserved by the same action. The specified capacity, shared output, and low-energy weak endpoints delimit the construction.

For static response, equal pion-flavor masses, and a uniform inversion-symmetric background, take the same real strength g_π at both endpoints and set $G = (m_\pi^2 + \nu^2 L)^{-1}$ and $D_i = (S_i - S_i^\dagger)/(2i\ell)$. The cross response is

$$\boxed{V_{AB} = -2g_\pi^2 m_\pi (\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B) \sum_{ij} \sigma_A^i \sigma_B^j \langle D_i \delta_{x_A}, G D_j \delta_{x_B} \rangle.} \tag{3.35}$$

The factor two comes from the two directions of cross return. Equation (3.34) also gives both self energies; their presence does not imply attraction in every channel. Static elimination requires external frequencies and endpoint strengths below the accessible mediator gap, with nucleon motion slow relative to the mediator response.

For $m_\pi/\nu \ll 1$, in a long-wavelength window away from lattice and boundary scales, let $c = \nu\ell$, $\kappa = m_\pi/c$, and $Y = e^{-\kappa r}/(4\pi c^2 r)$. The continuum point-source strength is $g_c = g_\pi \ell^{3/2}$. For $r > 0$, Eq. (3.35) becomes

$$\boxed{\begin{aligned} V_{AB}(r) &= \frac{2g_c^2 m_\pi}{3} (\boldsymbol{\tau}_A \cdot \boldsymbol{\tau}_B) Y(r) \\ &\quad \times \left[\kappa^2 \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B + \left(\kappa^2 + \frac{3\kappa}{r} + \frac{3}{r^2} \right) S_{12} \right], \\ S_{12} &= 3(\boldsymbol{\sigma}_A \cdot \widehat{\boldsymbol{r}})(\boldsymbol{\sigma}_B \cdot \widehat{\boldsymbol{r}}) - \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B. \end{aligned}} \tag{3.36}$$

The central and tensor terms follow jointly from the same endpoints and propagation and can be compared with the corresponding meson-exchange structures [21, 22]. The central term attracts for total isospin and spin $(I, S) = (1, 0), (0, 1)$, and repels for $(0, 0), (1, 1)$. The triplet tensor term further distinguishes directions and orbital components. Identical nucleons require fermionic antisymmetry. At the origin one must retain the finite kernel or the contact term of the same distribution, rather than extend the far-field expression to zero separation.

Section D5 supplies the complete nucleon intertwiner, finite kernel, contact term, and error bound. The result explains channel-dependent interaction in a nucleon–pion joint process. It does not yet include multipion nuclear forces, all short-range interactions, or nuclear bound-state spectra.

§4 Fields and Light

Light connects interactions to distant measurements. Emitting matter transfers energy and phase into propagating optical modes, from which receiving matter acquires a response; during propagation, changes in the background alter the phase, direction, and arrival time of light. Light therefore participates in both field propagation and field observation. Its zero rest mass, electrical neutrality, and role as a measurement signal arise, respectively, from the adopted modes, couplings, and recording processes.

§4.1 Light Propagation and Phase

Chapter 3 inherits the original photon zero mode from the neutral mass response and derives two transverse propagation components in the controlled continuum window of flux and loop return. Plane waves occur in a source-free uniform region. With $\hbar = c_0 = 1$, this continuum description gives

$$\boldsymbol{a}(\boldsymbol{x}, t) = \boldsymbol{\epsilon}\, e^{i(\boldsymbol{k}\cdot\boldsymbol{x} - \omega t)}, \qquad \boldsymbol{k}\cdot\boldsymbol{\epsilon} = 0, \qquad \omega = c_{\text{field}}|\boldsymbol{k}|. \tag{4.1}$$

A localized wave packet is a superposition of these components. Phase differences between neighboring wave numbers determine how the envelope moves, with group velocity $\boldsymbol{v}_g = \nabla_{\boldsymbol{k}}\omega$. The photon energy and momentum are calibrated in the same physical units as $E_\gamma = \omega$ and $\boldsymbol{p}_\gamma = \boldsymbol{k}$. Zero rest mass allows all of the energy to take the form of propagation energy; it does not eliminate either the energy or its action on receiving matter.

Equation (4.1) uses the declared continuum working window of Chapter 3. Reference units $c_0 = 1$ do not automatically set c_{field} to one; $c_{\text{field}} = c_0$ remains a matching condition with the original physical calibration. The phase of the actual discrete propagation is still given by the complete cycle, and the group velocity is calculated using the total elapsed duration; at finite wave number, the actual dispersion of the adopted shift action must be retained [2, 3]. The long-wavelength speed and frequency of light are calibrated together by this propagation.

The phases of matter and light can be compared using their respective continuum dispersions. Let c_s denote the actual speed coefficient of the branch under consideration. On the positive-energy branch $E = \sqrt{c_s^2|\boldsymbol{p}|^2 + m^2}$, $\boldsymbol{v}_g = c_s^2\boldsymbol{p}/E$. Along the trajectory of the wave-packet center, the rate of change of the plane-wave phase $S = \boldsymbol{p}\cdot\boldsymbol{x} - Et$ is

$$E = \boldsymbol{p}\cdot\boldsymbol{v}_g + \frac{m^2}{E}, \qquad \frac{dS}{dt} = -\frac{m^2}{E}. \tag{4.2}$$

The right-hand side gives the residual phase accumulation along this trajectory. Let τ_{p} be proper time in the continuum clock representation of this massive branch. At rest, this rate of change is $-m$; for moving massive matter, it equals $-m\, d\tau_{\text{p}}/dt$. Light satisfies $m = 0$, so this phase-change rate vanishes along a light ray, and the same wavefront extends with propagation. A detector at a fixed position still encounters successive wavefronts at frequency ω and can therefore measure the frequency of light. The phase change along a light ray is thus distinguished from the frequency reading of a fixed apparatus.

Contact between light and matter occurs at the emission and reception vertices. The motion of charged matter can excite electromagnetic modes, which in turn change the state and momentum of local matter upon reaching the receiver. Energy transfer in a complete process also includes recoil, changes in internal energy levels, and supply from the apparatus. Free propagation and the exchanges at both ends are connected within the same joint evolution, from whose final state the actual pointer reads out an outcome.

§4.2 Light Generation, Source Writing, and Response

Zero photon charge does not prevent charged matter from exciting the light field. Matter moving along an actual edge changes background flux. A loop return can restore matter position while leaving a divergence-free background excitation. A three-dimensional local curl deployment places the earliest output and the single-pattern response with returns in transverse propagation; the complete multiple-pattern action remains in the same model. Matter current, background generation, and reverse absorption all arise from one joint action.

One finite realization carries the source on two tree edges and closes the loop with a third. Write the tree source as H_{tree}, the closing edge as V, and contact plus distant transport as H_B. After preparing a releasable coherent combination in the original source spectrum, the complete action obeys

$$H = H_{\mathrm{tree}} + V + H_B, \qquad \frac{d}{dt}\langle H_{\mathrm{tree}} + V + H_B\rangle = 0. \tag{4.3}$$

The source spectrum and lowering matrix element follow from the original mass and edge strength. Work for phase preparation and restoration of control must be included. Amplitudes generated at different times combine within the same propagation. With $H_0 = H_{\mathrm{tree}} + H_B$,

$$|\Psi(t)\rangle = e^{-itH_0}|\Psi_0\rangle - i\int_0^t e^{-i(t-s)H_0} V|\Psi(s)\rangle\, ds. \tag{4.4}$$

This gives finite-time matching between source-lowering and propagation frequencies while retaining returns and source depletion. Section D1 supplies the loop matrices, actual source preparation, and positive earliest distant net energy output. That output includes the positive and negative frequencies and multiple patterns allowed by the original action. It is not thereby a pure single photon, and conservation of the activity generator does not replace the separate bare-shell energy accounting. The background-maintenance construction of Chapter 3 additionally gives electromagnetic storage and continuum response; the extra deployments of the two constructions are stated separately.

The gravitational background written by light or moving matter requires a different actual contact. Let K_s be the complete source generator coupled to a fresh auxiliary, with input ρ_s. The read–exchange process of Chapter 1 gives

$$b_s = \frac{1}{2}\,\mathrm{Tr}[\rho_s \sin(2t_{\mathrm{read}} K_s)], \qquad \Delta\xi_s = \varepsilon_s(b_s - \xi_s). \tag{4.5}$$

For a small read angle, $b_s = t_{\mathrm{read}}\langle K_s\rangle + O(t_{\mathrm{read}}^3 \|K_s\|^3)$. Coupling the complete motion generator reads E on a positive-energy input, rather than rest mass alone.

Zero mass does not make this action vanish. Meanwhile, the specified common clock–rod preparation of Chapter 2 has long-wavelength reception energy $E(A,p) = \sqrt{m^2A^2 + A^4|p|^2}$. Its conjugate response to availability at the field-free baseline satisfies

$$\left.\frac{\partial E(A,p)}{\partial A}\right|_{A=1} = \frac{m^2 + 2|p|^2}{E} = E + \boldsymbol{p}\cdot\boldsymbol{v}. \tag{4.6}$$

The writing in Eq. (4.5) and the reception contribution in Eq. (4.6) concern different observables. Persistent evolution also retains contributions from the background's own action. The conjugate response cannot be substituted for net occupation injection.

To retain source, transport, and target reception simultaneously, use the finite joint action

$$\begin{aligned} H_{\mathrm{tot}} &= K_s + H_B + V_w + H_{\mathrm{tr}} + H_C, \\ \frac{d}{dt}\langle K_s + H_B + V_w + H_{\mathrm{tr}} + H_C\rangle &= 0. \end{aligned} \tag{4.7}$$

Changes in every term follow from commutators at the same time, with source–background correlations retained. Section D7 proves a bound for the time-ordered mean response in an evolving environment and displays occupation covariance and exchange coherence in the three-package coefficient. A single mean occupation does not close a general persistent process.

A three-dimensional local light source must also respect the propagation constraint. If K_γ is the original transverse propagation and P_x a position projector, the direct action $\{P_x, K_\gamma\}/2$ generally fails to preserve bare transversality. For a fixed positive energy scale $\Lambda > 0$ independent of the input energy, the two-sided curl action $K_\gamma P_x K_\gamma/\Lambda$ preserves it but sums to K_γ^2/Λ, changing the source spectrum to E^2/Λ. Thus the finite mechanism for a complete moving source is established, while arbitrary three-dimensional local light writing, control of persistent correlations, and a universal strong-field constitutive relation still require a common realization.

§4.3 Free Radiation and Composite Sources

Freely propagating radiation and radiation confined within matter participate in different complete processes. The components of a free stream propagate individually; when radiation is confined within a composite system, it repeatedly exchanges momentum and energy with the boundary or internal matter, and the matter maintaining the confinement also participates in the overall response. The object treated as the source must be consistent with this complete process.

In a local continuum representation, describe a free particle stream by a distribution function $f(\boldsymbol{x},\boldsymbol{p})$, and let $d\Gamma_p$ be the momentum measure with the same normalization convention. Its energy density and spatial stress are

$$\rho_E = \int d\Gamma_p\, fE, \qquad \Pi^{ij} = \int d\Gamma_p\, f\frac{p^i p^j}{E}. \tag{4.8}$$

Free light satisfies $\operatorname{tr}\Pi = \rho_E$: it transfers momentum as well as energy. When radiation is confined by a boundary or by internal matter, its stress must be accounted for together with the action of the confining matter.

The rest mass of composite matter is read from its total energy. If the complete state at rest satisfies $H_C|\Psi_C\rangle = M|\Psi_C\rangle$, then H_C includes both the interaction energy and the matter providing the constraints. Under the source calibration of the total generator in Chapter 2, it is M that enters the source relation at rest. For example, after a closed container absorbs and retains radiation, its mass change, evaluated in the same rest reference frame, equals the net retained energy; the energy balances of the emitting apparatus and the surroundings are accounted for separately. The zero rest mass of the internal photons does not eliminate the increase in total energy.

In a flat continuum representation that neglects self-gravity, if the complete system has a stress–energy tensor satisfying $\partial_\mu T^{\mu\nu} = 0$, stationarity in the rest frame provides a useful balance relation. For a localized system at rest, with no external support and a time-independent stress–energy tensor, $\partial_k T^{kj} = 0$, and the surface term at infinity vanishes. Integrating $\partial_k(x^i T^{kj})$ gives

$$\int d^3x\, T^{ij} = 0, \qquad \int d^3x\, T^{00} = M. \tag{4.9}$$

This integral relation is Laue's theorem for a system at rest [23]. The total stress includes internal radiation, interactions, and confining matter. The pressure of the internal light is balanced together with the stresses of the container or binding structure, while the total rest mass is determined by the complete energy.

The propagation energy of free radiation, the rest mass of a composite system, and the actual source writing thus each refer to a definite object. Which matter and exchanges are included in a system, and which boundaries provide its constraints, determine how its source should be accounted for.

§4.4 Light, Clocks, and Length Scales

As light propagates through the background, it connects the calibrations of time and length at different positions. When light and matter share the static isotropic geometry of Section 2.5, $ds^2 = A^2dt^2 - B^2d\boldsymbol{x}^2$. In the geometric-optics regime [24], write the phase as S, with $k_\mu = \partial_\mu S$ denoting its gradient. Massless propagation satisfies

$$g^{\mu\nu}k_\mu k_\nu = 0, \qquad E_\infty = -\partial_t S, \qquad \omega_{\mathrm{loc}} = \frac{E_\infty}{A}, \quad |\boldsymbol{k}|_{\mathrm{loc}} = \frac{|\nabla S|}{B}. \tag{4.10}$$

Here E_∞ is the frequency relative to the distant reference time; in a static background, it is conserved along propagation without additional frequency conversion. A local apparatus at rest reads out ω_{loc} using its own time and the wave number using its local length scale, with $\omega_{\mathrm{loc}} = |\boldsymbol{k}|_{\mathrm{loc}}$. The local speed of light is therefore one, whereas its coordinate speed is A/B. The two readings use different scales.

This makes the redshift relation of Chapter 2 a direct signal comparison. The local frequencies at emission and reception are E_∞/A_e and E_∞/A_r, respectively, so $\omega_r/\omega_e = A_e/A_r$. When a signal travels from a region of lower clock rate to one of higher clock rate, the receiver reads a lower frequency. Successive wavefronts maintain the same coordinate-time interval, while the clocks at the two ends assign different local durations to that interval.

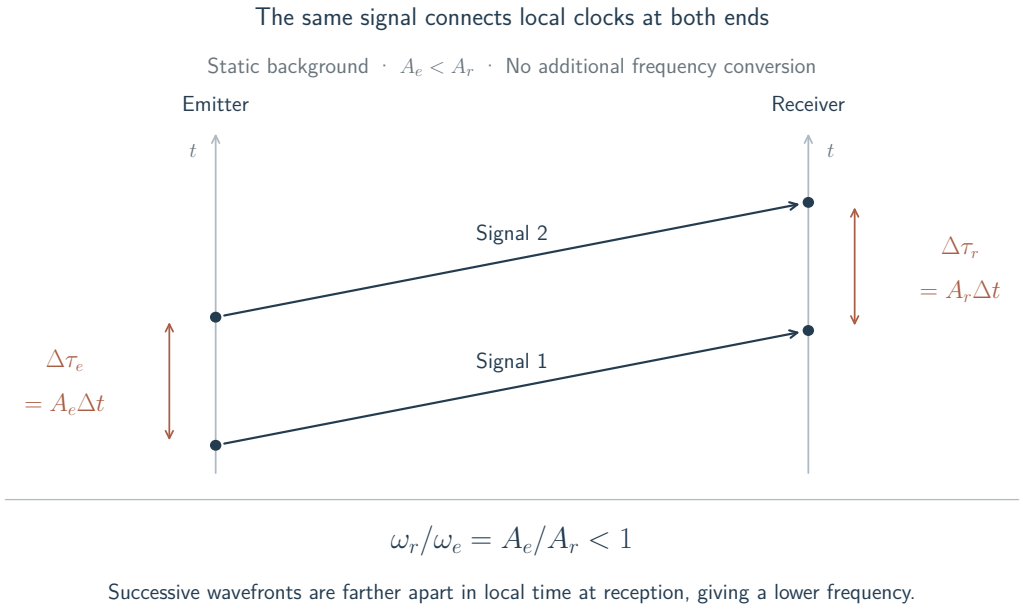


Figure 4.1. Light signals connect clock comparisons. In a static background, two emissions are separated by the same reference duration Δt, and the corresponding receptions are also separated by Δt. The local clocks at the two ends record $A_e\Delta t$ and $A_r\Delta t$, respectively, yielding the frequency ratio A_e/A_r. The slanted lines indicate only the signal connection from earlier emission to later reception; their slopes do not specify the coordinate speed along the path.

The arrival time of light also reads out the spatial length scale. The null-interval condition gives

$$dt = n_{\mathrm{opt}}(\boldsymbol{x})\, d\ell, \qquad n_{\mathrm{opt}} = \frac{B}{A}, \qquad \mathcal{T}_\gamma = \int_\gamma n_{\mathrm{opt}}\, d\ell. \tag{4.11}$$

Here $d\ell = |d\boldsymbol{x}|$, and $\mathcal{T}_\gamma$ is the reference propagation time along the path. The optical refractive index n_{opt} is given by the relation between the time and length scales. Light paths vary according to this optical length, and a difference in propagation time between two paths can be converted into a phase difference. Round-trip signals connect emission, reflection, and reception into a distance measurement; gravitational time delay can be tested through such round-trip timing [25].

For static propagation between the same fixed endpoints, a small change in the optical refractive index produces the following leading change in propagation time along the zeroth-order stationary light path γ:

$$\delta\mathcal{T} = \int_\gamma \delta n_{\mathrm{opt}}\, d\ell, \qquad \delta\varphi = E_\infty\, \delta\mathcal{T}. \tag{4.12}$$

Stationarity with fixed endpoints ensures that the path variation contributes no additional first-order optical-path term. The second equation compares monochromatic propagation phases at fixed distant frequency E_∞ and a common detection time; a comparison of wave-packet arrival times reads out the first equation directly. Propagation time and deflection therefore both read out spatial changes in B/A, with their specific magnitudes determined by the actual propagation action.

Local rod length, remote round-trip distance, and coordinate separation are each determined by their corresponding operations. In a sufficiently small neighborhood, round-trip timing calibrates length using the local speed of light; across a region with a changing clock rate, the path integral in Eq. (4.11) must be retained. The emitting and receiving apparatuses, the phase of light, and the background remain within the same joint state. Observational spacetime organizes these internal processes into comparable distances and times.

§4.5 Optical Recording

Upon reaching the receiver, light changes the apparatus through an actual coupling. Energy absorption, scattering, polarization selection, and phase comparison can correspond to different recording schemes; which distinctions the apparatus retains is determined by its pointer and the subsequent process that preserves the record. The record construction in Ref. [2] allows general auxiliary degrees of freedom to serve as these pointers. Light is an important carrier of signals and energy between separated positions, while a specific record is formed jointly by the light and the apparatus.

For a normalized optical input state ρ_γ, let $M_r^\dagger M_r$ denote the outcome effect determined by the fixed apparatus preparation and contact process. Including the no-detection outcome, the outcomes satisfy [5]

$$p_r = \mathrm{Tr}(\rho_\gamma M_r^\dagger M_r), \qquad \sum_r M_r^\dagger M_r = I. \tag{4.13}$$

The detection probability therefore includes the matching between the incident mode and the apparatus, the coupling duration, the acceptance range, and the detection efficiency [26]. The electromagnetic coupling enters the contact amplitude; its numerical value is not itself the probability of producing a record in each cycle. Changes in the energy and momentum of the receiver are also read from the same output.

Phase is recorded through comparison. Consider an ideal interferometer with equal-weight beam splitting and recombination, balanced losses in the two arms, and identical detection efficiencies at the two outputs. Let the two histories accumulate a phase difference $\Delta\varphi$ at the receiver, and include the phase of the environment overlap in θ_{env}. The conditional probabilities of the two complementary outputs are then

$$p_\pm = \frac{1}{2}\left[1 \pm \mathcal{V}\cos(\Delta\varphi + \theta_{\mathrm{env}})\right], \qquad \mathcal{V} = |\langle e_2|e_1\rangle| \le 1. \tag{4.14}$$

Here the probabilities have been normalized to the light that successfully arrives; actual losses remain included in Eq. (4.13). The field changes the phase along the paths, and the interferometer converts this change into a change in the probabilities of the two outcomes. When the environment retains a distinction between the paths, $\mathcal{V}$ decreases, changing the extent to which the phase information can be read out. The detected event and the preceding coherent propagation are connected within the same process.

Repeated records form statistical readings of the field. Let X_a denote a counting outcome in trial a, and let $N = \sum_a X_a$. Then

$$\operatorname{Var} N = \sum_{a} \operatorname{Var} X_a + 2 \sum_{a<b} \operatorname{Cov}(X_a, X_b). \tag{4.15}$$

Correlations in the source and apparatus enter the second term. Independent, identically distributed counts give the usual square-root fluctuation scaling; when correlations are present, the actual covariances must be used. The intensity, phase, and polarization of light can thus be reconstructed through various statistics [26, 27]; readings of a continuous field are formed from actual events and their correlations.

§5 Field Phenomena

The spatial distribution of a field manifests itself through matter motion, energy exchange, and instrument records. The action on distant matter must pass through background propagation; the response produced by light at the receiver must pass through the actual coupling between light and the apparatus; and continuously varying field quantities are read out through the apparatus's comparisons and statistics of these processes.

§5.1 Long-Range Action

A persistent source can maintain a measurable action at a distant position. Immediately after the source changes, however, the distant region does not acquire the new stationary value. The stationary distribution describes the spatial relation after the persistent action has been established, while the build-up process retains the time required for each segment of propagation.

The background recurrence of Chapter 2 directly connects these two aspects. Fix the refreshing and transport rules, and let $T = R(I - D_\varepsilon)$, $q_n = RD_\varepsilon b_n$. Comparing two source histories with the same initial state gives

$$\delta\xi_N = \sum_{n=0}^{N-1} T^{N-1-n}\delta q_n. \tag{5.1}$$

Each source change first forms δq_n in the current cycle and then undergoes propagation over the remaining cycles. For an implementation that transports by at most one neighboring position per cycle, $T_{ij} = 0$ whenever positions i, j are neither identical nor adjacent. Denoting the shortest number of steps on this actual exchange graph by $d(i, j)$, we have

$$(T^k)_{ij} = 0 \qquad \text{when } d(i, j) > k. \tag{5.2}$$

This follows directly from a finite number of local exchanges: before the required number of steps has elapsed, no sequence of actions has yet carried the change at that position to the receiver. The starting point in Eq. (5.1) is the support of the current-cycle output q_n, which already includes the transport step performed after source writing. The number of steps here is converted into elapsed duration using the actual exchanges and complete timing.

If the boundary supply makes propagation in this finite system decay, the long-term action of a constant source gives

$$\xi_* = (I - T)^{-1}q = \sum_{k=0}^{\infty} T^k q. \tag{5.3}$$

The long-range stationary response accumulates over many cycles of propagation. The three-dimensional far-field $1/r$ kernel of Chapter 2 is precisely the long-term distribution obtained in the corresponding spatial structure; it does not remove the temporal ordering in the finite-duration solution.

A coherent field propagates according to its own evolution equation. For the linear electromagnetic field of Chapter 3, in Lorenz gauge and with boundary con-

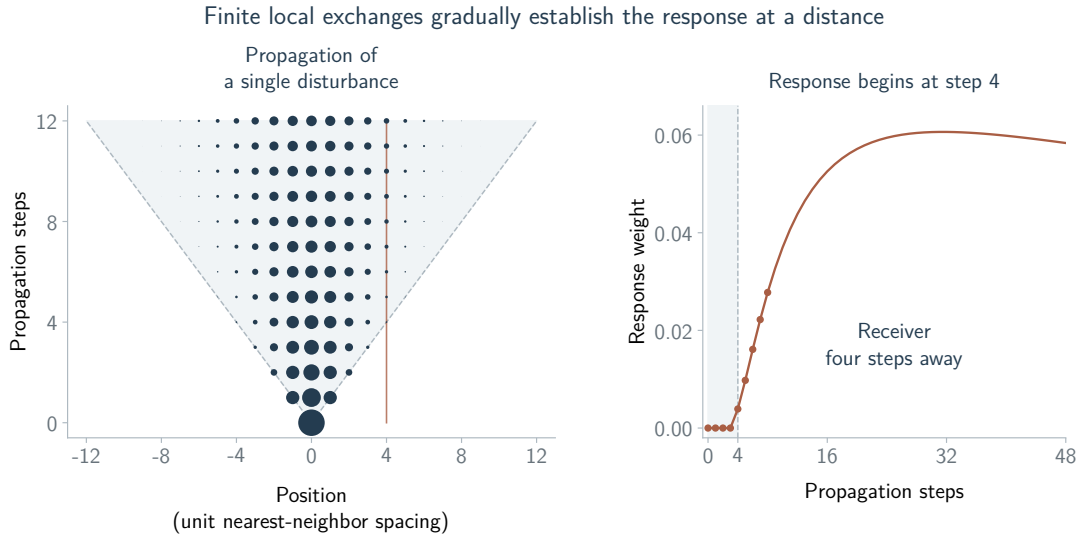


Figure 5.1. A distant response undergoes successive local propagation. The left panel starts with a unit perturbation at the center of a one-dimensional chain. In each cycle, it remains at the same position with weight 1/2 and is transported to either side with weight 1/4. The area of each dot represents the response weight, and the dashed lines mark the range reached by traversing at most one neighboring position per cycle. The right panel reads out a unit four steps away: there is no response before step 4, after which different propagation histories accumulate a nonzero weight. This is a finite-step example of Eq. (5.2).

ditions excluding any additional incident field, the retarded solution for a conserved current is

$$\begin{aligned} \phi(t,\boldsymbol{x}) &= \int d^3x'\, \frac{\rho(t-R/c,\boldsymbol{x}')}{4\pi\varepsilon R}, \\ \boldsymbol{A}(t,\boldsymbol{x}) &= \mu \int d^3x'\, \frac{\boldsymbol{j}(t-R/c,\boldsymbol{x}')}{4\pi R}, \qquad R=|\boldsymbol{x}-\boldsymbol{x}'|. \end{aligned} \tag{5.4}$$

Here the fields and coefficients are physical quantities measured in actual reference time, with $c = c_{\text{field}} = 1/\sqrt{\varepsilon\mu}$. Each contribution reaching the receiver at time t is evaluated at the corresponding retarded source time. Once the source and current cease to change, this reduces to the static kernel. The diffusion description of occupation and this coherent-wave description represent different responses; similarity in their static forms does not by itself determine their dispersion or propagation speeds. The temporal structure of long-range action is always given by the actual propagation.

§5.2 The Photoelectric Effect

When light illuminates matter, the incident energy can become internal excitation, heat, or scattered light, and it can also cause electrons to escape from the surface. The photoelectric effect corresponds to the electron-emission channel: the material absorbs light, an electron overcomes the energy difference required to escape its binding, and the collection apparatus produces a current or an individual count.

The energy transferred in a single event is determined by the joint action. The energy-conserving exchange in Ref. [2] gives an explicit expression. If the incoming and outgoing auxiliary states have energies $\varepsilon_i, \varepsilon_o$, and the corresponding matter transition operator is J_{oi}, then

$$[H_S, J_{oi}] = (\varepsilon_i - \varepsilon_o) J_{oi}. \tag{5.5}$$

This equates the energy lost by the auxiliary with the energy gained by matter. For optical absorption, the incident photon has energy $\hbar\omega$, and this change is jointly accommodated by the material, the electron, and recoil. To express the relation between frequency and energy conveniently, this section writes $\hbar$ explicitly.

In the weak-light regime dominated by single-photon absorption, define the material's work function W by the lowest-energy escape channel. Then

$$\hbar\omega = W + K_e + E_{\text{extra}}, \qquad E_{\text{extra}} \geq 0, \qquad K_{\max} \simeq \hbar\omega - W. \tag{5.6}$$

Here E_{extra} includes additional excitation, recoil, and losses relative to the lowest-energy channel. The last relation takes the upper edge of the kinetic energy where these additional costs are negligible, recovering Einstein's photoelectric relation [28]. When the single-photon energy is below W, this escape channel is closed, although the material may still absorb or scatter light through other channels. Above threshold, the frequency determines how much energy a single absorption can provide.

The light intensity I is the energy arriving per unit time and unit area. For incident light of fixed frequency illuminating an area S uniformly for a duration $\mathcal{T}$, let η be the efficiency with which an incident photon produces one collected primary photoelectron. In the single-photon regime, with the material state and efficiency maintained and the apparatus not yet saturated,

$$\langle N_e \rangle = \eta \frac{IS\mathcal{T}}{\hbar\omega}. \tag{5.7}$$

Increasing the light intensity therefore primarily increases the number of collected electrons, whereas changing the frequency changes the energy available in a single event and determines whether the material's emission threshold is crossed. The efficiency η is given by the actual absorption, electron transport, and collection processes and may also depend on frequency; it connects to the detection apparatus of Chapter 4.

The apparatus can reduce its output to "detected" and "not detected," or it can further resolve the electron's energy and direction. A binary outcome can combine many different energy transfers. The energy transferred is determined by the actual transition, while the binary character specifies only which distinction the apparatus retains. The amplitudes of continuous propagation determine the probabilities of the channels, while individual absorption and subsequent recording produce countable events.

§5.3 Energy Transfer by Fields

A source loses energy when it emits radiation, receiving matter gains energy when it absorbs radiation, and the intervening propagation belongs to the same process. The common background, propagating excitations, interactions, and auxiliaries involved

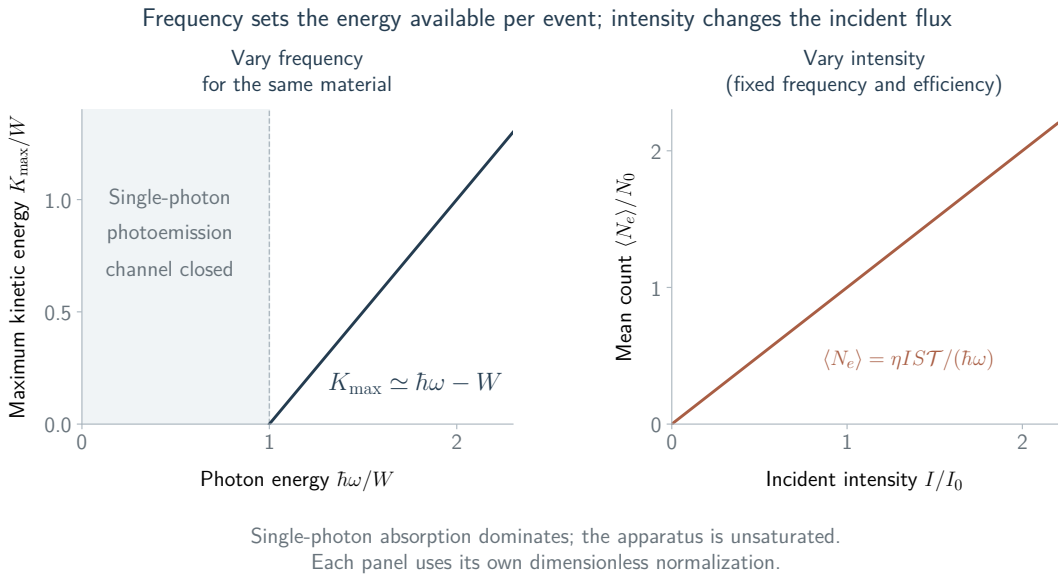


Figure 5.2. Frequency and light intensity affect different outputs. The left panel shows the ideal upper edge of the kinetic energy in Eq. (5.6): this single-photon escape channel is absent in the region $\hbar\omega/W < 1$. The right panel shows how the count in Eq. (5.7) varies with light intensity at fixed frequency and efficiency. The curves are normalized by an arbitrary positive reference intensity I_0 and its count N_0, without introducing specific material parameters; multiphoton processes and saturation lie outside the scope of these two relations.

in preparation and retrieval all enter the complete Hamiltonian. Denoting this Hamiltonian by $H_{\mathrm{tot}}(t)$, we have

$$\frac{d}{dt}\langle H_{\mathrm{tot}}\rangle = \left\langle \frac{\partial H_{\mathrm{tot}}}{\partial t} \right\rangle . \tag{5.8}$$

For a complete closed system with no external time-dependent control, the right-hand side vanishes. If prescribed switching or driving is used, the right-hand side is the corresponding work rate; once the driving apparatus is included in a larger closed system, this exchange also becomes part of joint energy conservation. The supply required for source maintenance and background refreshing therefore cannot be omitted from the energy balance.

Local energy flux describes how energy crosses the boundary of a region. The electromagnetic model of Chapter 3 gives $u_{\mathrm{em}} = \varepsilon|\boldsymbol{E}|^2/2 + |\boldsymbol{B}|^2/(2\mu)$ and $\boldsymbol{S}_{\mathrm{em}} = \boldsymbol{E}\times\boldsymbol{B}/\mu$. All quantities here use physical readouts and coefficients in actual time. Integrating over a fixed region V yields

$$\frac{d}{dt}\int_V u_{\mathrm{em}}\, d^3x = -\oint_{\partial V} \boldsymbol{S}_{\mathrm{em}}\cdot d\boldsymbol{a} - \int_V \boldsymbol{j}\cdot\boldsymbol{E}\, d^3x. \tag{5.9}$$

Field energy within the region can increase through inflow across the boundary and decrease through outflow or work done on matter. The energy acquired by the receiving material and apparatus is connected to the field through the final term. Even when no detector is placed along the propagation path, the field state and these energy flows are still given by the evolution; recording determines which quantities the apparatus retains, not whether the propagating energy exists.

For other fields, local energy and energy flux must be determined from the corresponding actions. The finite net energy of directional radiation is calculated in Supplement D8; persistent far-field power and positive-definite macroscopic energy still require the same propagation and actual material identification. Conservation for the full system specifies the total balance, while the specific distribution is determined by the propagation dynamics.

§5.4 Directional Radiation and Interferometric Observation

The directional action of matter can change background shear response. Free radiation additionally requires an actual propagation carrier, a source action, and constraints preserved by both. We consider an explicit conditional construction, distinguishing absorption of radiation from its connection to a geometric change in the original material clocks and rods.

A symmetric tensor amplitude $T \in \mathrm{Sym}^2(\mathbb{C}^3)$ carries six directional components, alongside a separate unexcited state and two propagation arms. It differs from the density readout of a single direction package. For centered-difference symbol $d_i = \sin(k_i \ell)/\ell$, set $C = i[d]_\times$ and $\mathcal{L}T = (CT + TC^{\mathsf{T}})/2$. Take the local propagation and source operator

$$\boxed{h_g = i c_g \begin{pmatrix} 0 & \mathcal{L} \\ -\mathcal{L} & 0 \end{pmatrix}, \qquad \mathcal{B}_3(J) = \mathcal{L}(CJC^{\mathsf{T}}).} \tag{5.10}$$

The three-difference combination satisfies $\mathrm{Div}\, \mathcal{B}_3 = 0$ and $\mathrm{tr}\, \mathcal{B}_3 = 0$. Coupling it to the actual energy-lowering block of matter, while retaining adjoint return and a receiver of the same type, makes source, propagation, and absorption jointly preserve the transverse traceless constraints. For $|d| \neq 0$, the constrained positive-frequency space has dimension two. Other free modes and negative-frequency branches remain in the complete background. Preparing this invariant sector does not mean evolution attracts arbitrary inputs into it.

The same finite action can produce positive short-time net field energy, excite a receiver, and distinguish polarization. Each actual port carries a $|d|^3$ filter, which cannot be divided out at emission or reception. Section D8 gives the full free spectrum, two-end response, and energy balance. Persistent far-field power, positive-definite macroscopic radiation energy, and a common calibration of c_g with light speed have not yet been established.

The material connection imposes an additional requirement. The natural two-index lift of the original directional readout gives $\widetilde{q}(T) = \mathrm{Re}(TT^\dagger) - \|T\|^2 I/3$. At fixed propagation direction $n = d/|d|$, every transverse traceless amplitude satisfies

$$\widetilde{q}(T) = \|T\|^2 \left(\frac{I}{6} - \frac{nn^{\mathsf{T}}}{2} \right). \tag{5.11}$$

This reads only intensity in that direction and cannot distinguish equally intense linear polarizations. The actual original material contact also generates trace or longitudinal components. A coherent reference can provide a first-order readout under a specified preparation, but its formation and constraint-preserving service have not been completed. The radiation candidate therefore cannot directly be identified with a first-order geometric perturbation of the original clocks and rods.

Interferometry describes the subsequent readout: once complete matter and light propagation determine the optical-path changes of both arms, the returning phase difference becomes a detector output [29]. This relation can be stated independently, but does not replace the dynamical connection from directional radiation to actual optical paths.

To illustrate this relation, consider an ideal Michelson interferometer with two equal-length arms, each completing one round trip. In the regime where the perturbation varies slowly compared with the round-trip time, let $\delta L_1, \delta L_2$ denote the equivalent changes in the one-way optical lengths of the two arms, and set $\Delta L = \delta L_1 - \delta L_2$. Calibrating the laser wavelength λ and ΔL with the same reference time and length scales, and keeping the laser frequency fixed, gives

$$\delta\varphi = 2k\Delta L = \frac{4\pi}{\lambda}\Delta L, \qquad k = \frac{2\pi}{\lambda}. \tag{5.12}$$

The round trip in each arm doubles the optical-path change relative to a one-way passage. Here ΔL is obtained from the complete light paths and the response of the test bodies. For faster variations, the actual round-trip histories must be integrated, retaining time delays and the frequency response of the instrument. The static optical-path relation of Chapter 4 is the static case of this description.

At the point of maximum slope of the interference output in Eq. (4.14), fix the visibility at $\mathcal{V} > 0$ and estimate a small phase change from N independent, successfully detected outcomes. At this point, $p_+ = 1/2$, the standard deviation of its count fraction is $1/(2\sqrt{N})$, and the magnitude of the phase slope is $\mathcal{V}/2$. Thus

$$\sigma_\varphi \simeq \frac{1}{\mathcal{V}\sqrt{N}}, \qquad \sigma_{\Delta L} \simeq \frac{\lambda}{4\pi\mathcal{V}\sqrt{N}}. \tag{5.13}$$

This gives the counting-noise term for an ideal single-round-trip apparatus [30]. Multiple round trips, optical cavities, noise in the source and apparatus, and correlated states of light change the actual response and uncertainty. A smaller perturbation produces a smaller change in the output probabilities; accumulating many records allows this change to be distinguished from statistical fluctuations.

The continuous motion of interference fringes describes a change in a statistical distribution, while photon counting describes the absorption processes used to read it out. These two readings belong to different stages of the same measurement chain.

§5.5 Formation of Classical Fields

Macroscopic field readings are stable because the apparatus produces a reproducible response over a certain spatial and temporal range and estimates its mean quantities through many records. A continuum representation of the field further requires sufficiently smooth spatial variation within that range. Statistical stability and spatial smoothness together allow an instrument of finite resolution to describe the field by a continuous function.

Consider the statistics first. Let X_a be a real-valued reading in a sequence of records, with a common mean, variance $\sigma^2 > 0$, and lag correlation coefficients ρ_k under a stationary preparation. Averaging M records, Eq. (4.15) can be rearranged as

$$\operatorname{Var} \bar{X}_M = \frac{\sigma^2}{M} \left[1 + 2 \sum_{k=1}^{M-1} \left(1 - \frac{k}{M} \right) \rho_k \right]. \tag{5.14}$$

Independent records make the bracket equal to one, and the uncertainty of the mean decreases as $M^{-1/2}$. The same leading scaling holds when the correlation tail is summable and the asymptotic variance coefficient is nonzero; if the source or matter retains long-term memory, the correlation terms must be estimated together. Replacing the auxiliary with a fresh one in every cycle specifies only the preparation of the fresh auxiliary; it does not erase correlations retained in the matter and the existing background.

Now consider space. A continuum equation approximates a specific discrete propagation. For example, for a uniform one-dimensional centered difference with spacing ℓ, a sufficiently smooth field quantity f satisfies

$$\frac{f(x+\ell) - 2f(x) + f(x-\ell)}{\ell^2} = f''(x) + \frac{\ell^2}{12} f^{(4)}(x) + O(\ell^4). \tag{5.15}$$

When the scale of field variation is much larger than ℓ, the leading differential term describes spatial variation on larger scales, and higher derivatives provide corrections. Source points, boundaries, and sharply varying interfaces still require the corresponding local solutions. This expansion is determined by the discrete operator and the smoothness of the field; its scale cannot be assigned solely from a cycle frequency.

After a mean has been read out stably, the equation it obeys is still determined by the dynamics. The refreshing recurrence with fixed coefficients in Chapter 2 and the linear Maxwell principal part obtained in the controlled window of Chapter 3 allow averaging to commute with the linear action, so the corresponding mean quantities evolve under the same linear equations. Nonlinear actions retain contributions from fluctuations. For example, for two classical statistical field quantities F, G,

$$\langle FG \rangle = \langle F \rangle \langle G \rangle + \langle \delta F \, \delta G \rangle. \tag{5.16}$$

Increasing the number of observations allows the terms on the right-hand side to be estimated more accurately, but does not automatically eliminate the correlation term. In the regime where fluctuations are small relative to a slowly varying background and correlation terms can be controlled, the mean field can approximately form a closed classical description; if correlations still play a significant role, the corresponding correlation quantities must also be retained.

The continuum equations of classical fields are determined by these averaging processes, and their accuracy depends on statistical fluctuations, spatial variation scales, and the degree of mean-field closure.

§6 Source Response and Tests

Source writing determines the local input, while propagation and return determine how this input accumulates and builds up. Comparing finite readout, stationary distributions, and temporal responses allows contact and transport to be examined separately and tests whether they belong to the same process.

§6.1 Finite Source Readout

Consider the fixed eigenstate source of Chapter 2, keeping the readout procedure, update probability, and supply boundary unchanged. Let $\chi = 2t_{\mathrm{read}}\lambda_s$. Equations (1.2) and (2.4) give $b_s = \sin\chi/2$ and the actual net source j_s. Define the linear reference by the zero-phase slope of the same process, $j_{s,\mathrm{lin}} = \chi(\partial j_s/\partial\chi)_{\chi=0}$. Then

$$\frac{j_s}{j_{s,\mathrm{lin}}} = \frac{\sin\chi}{\chi} = 1 - \frac{\chi^2}{6} + O(\chi^4). \tag{6.1}$$

This ratio also applies to the far-field amplitude at the same position. The return kernel and refreshing coefficient remain in both outputs and cancel in the ratio; if the supply boundary or update probability changes, their effect on the absolute amplitude is still given by Eq. (2.4). The sinusoidal relation is the full output of this bounded readout and is not monotonic over the full phase range. This relation tests the finite-readout correction for this source contact. The output for a general source spectral distribution is obtained from Eq. (4.5).

§6.2 Build-up and Periodic Response

The temporal response uses the same propagation matrix T. Suppose a constant source changes at some instant and thereafter maintains the new q. The deviation from the new stationary state is

$$\xi_N - \xi_* = T^N(\xi_0 - \xi_*). \tag{6.2}$$

This equation describes both build-up and restoration. If a resolvable mode has a propagation eigenvalue $0 < \zeta < 1$ and each complete cycle has elapsed duration τ, it decays as ζ^N, corresponding to a relaxation time $-\tau/\log\zeta$. The static accumulation factor of the same mode under persistent input is $1/(1-\zeta)$. Greater accumulation and slower restoration are thus linked through the same eigenvalue.

Periodic modulation allows both amplitude and delay to be read out. For $\xi_{n+1} = T\xi_n + q_n$, take $q_n = \mathrm{Re}(q_\omega e^{-i\omega n\tau})$. The stable periodic response $\xi_n = \mathrm{Re}(X_\omega e^{-i\omega n\tau})$ satisfies

$$X_\omega = (e^{-i\omega\tau}I - T)^{-1}q_\omega. \tag{6.3}$$

At zero frequency, this reduces to the stationary response $(I-T)^{-1}$; as the frequency increases, the response amplitude and phase begin to resolve the temporal structure of propagation. For the single mode above, denote the ratio of the complex response amplitude to the complex source-input amplitude by $\mathcal{H}_\zeta(\omega)$. Then

$$\mathcal{H}_\zeta(\omega) = \frac{1}{e^{-i\omega\tau} - \zeta}, \qquad \left|\frac{\mathcal{H}_\zeta(\omega)}{\mathcal{H}_\zeta(0)}\right| = \frac{1-\zeta}{\sqrt{(1-\zeta)^2 + 4\zeta\sin^2(\omega\tau/2)}}. \tag{6.4}$$

A stationary measurement obtains only the zero-frequency response; an additional measurement of the build-up curve or periodic phase can test the propagation process that produces this distribution. An actual probe generally reads out multiple modes simultaneously, with weights determined by the source position, input procedure, and receiving coupling. What is tested here is the relaxation and accumulation of occupation transport.

A test can first determine the linear slope through small-phase readout and then compare the output at finite phase. Alternatively, one set of time-domain or frequency-domain responses can determine the propagation parameters, which are then used to calculate another set of responses. The readout phase and the total elapsed duration of each cycle require independent calibration, while the source and probe couplings remain consistent. In this way, the constraints imposed by the same propagation operator across different observations become relations that can be checked.

§7 Conclusions

A field is the distribution of background states and correlations formed through the joint exchange of matter and background. Source matter changes the background, local propagation carries this change to other positions, and receiving matter acquires changes in phase, energy, and momentum through its own coupling. The source output, field propagation, and action on matter thus belong to the same physical process; back-reaction, return, and energy exchange on all sides remain in the complete joint evolution.

Static and time-dependent fields are two manifestations of this process. For the occupation update adopted here, let $T = R(I - D_\varepsilon)$ and $q_n = RD_\varepsilon b_n$. Propagation in each cycle satisfies $\xi_{n+1} = T\xi_n + q_n$. When the specified supply boundary makes propagation decay, the response to differences in source history from the same initial state and the stationary distribution under constant input are, respectively,

$$\boxed{\begin{aligned} \delta\xi_N &= \sum_{n=0}^{N-1} T^{N-1-n}\delta q_n, \\ \xi_* &= (I - T)^{-1} q. \end{aligned}}$$

The first expression retains the temporal order in which local actions arrive through successive stages, while the second gives the spatial accumulation of persistent action. After the field becomes stationary, source writing, background transport, and replenishment at the boundary continue. The far-field response of the three-dimensional, equally weighted, nearest-neighbor conservative exchange used here has the form $1/r$, with its amplitude jointly determined by the net source output and return feedback. Actual propagation links the static profile to its build-up time.

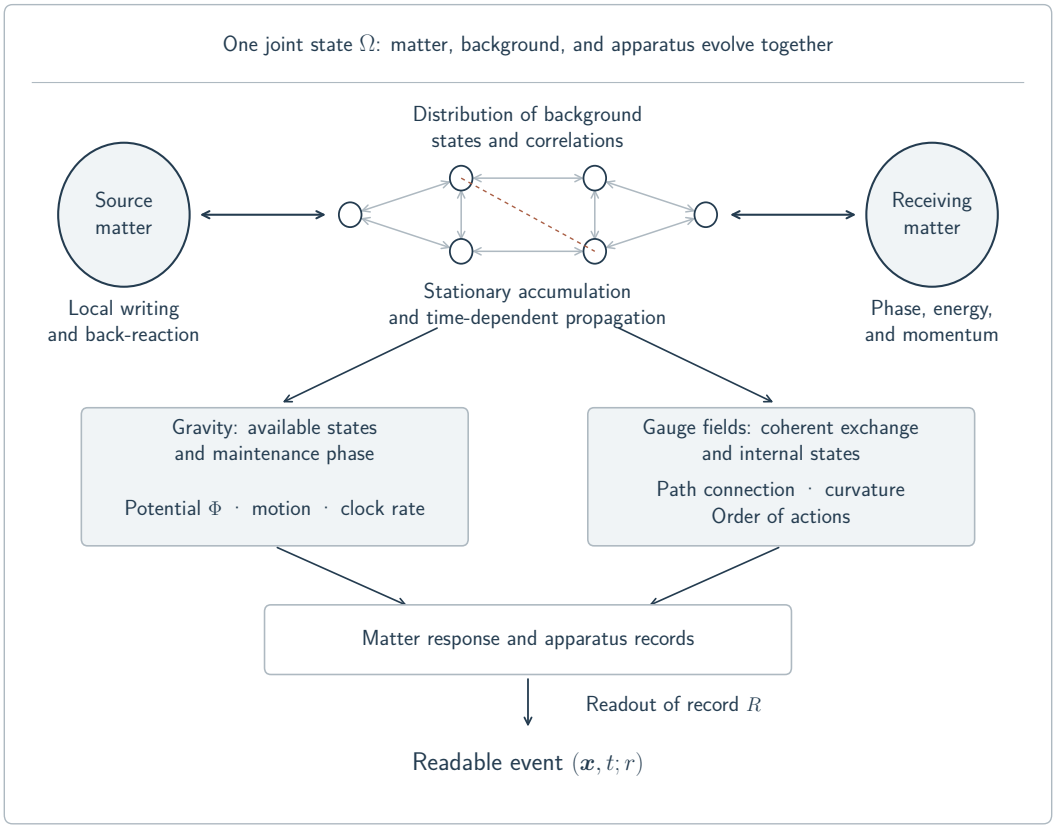


Figure 7.1. Field formation and observation belong to the same physical process. The source, background, receiving matter, and apparatus evolve jointly. The available background states and coherent transfer affect matter through the maintenance phase and internal exchange, respectively; the two response branches in the figure belong to the same joint state and do not represent two independent dynamics. The action leaves records, and readout produces events, with the record R placed on the readout arrow; event positions and times use a common $3+1$ representation.

Gravity links the available share of the background to the maintenance phase of matter. After a single physical calibration of the common strength, the external far field of a weak point source at rest, together with low-speed motion and clock rates in a slowly varying weak field, satisfies at leading order

$$\boxed{\begin{aligned} \Phi(r) &= -\frac{GM_s}{r}, \\ \ddot{\boldsymbol{X}} &= -\nabla\Phi, \\ d\tau_{\mathrm{loc}} &= (1+\Phi)\,dt. \end{aligned}}$$

The potential energy $m\Phi$ and kinetic energy $|p|^2/(2m)$ use the same mass coefficient, so the test mass cancels from the leading acceleration; the accumulation rate of the same phase also determines the local clock rate. Force and redshift thereby read the spatial gradient and temporal rhythm of the same background response. The same complete package reception further distinguishes single-package maintenance from three-package spatial overlap. Under the stated independent two-state occupation preparation, actual parallel reception, and finite-time control, $A = h/h_0$ and

$C_m = A^2$, hence $B = A/C_m = A^{-1}$. Persistent sources and transport produce further correlations, so this conditional relation is not a law for arbitrary backgrounds.

Electromagnetic storage and propagation follow from the same edge-flux and loop return. In the declared isolated upper band, actual auxiliary action gives

$$\boxed{H_{2,\mathrm{EM}} = H_m + \frac{a}{2}\sum_e N_e^2 + \frac{b}{2}\sum_p (I - W_p)^\dagger (I - W_p).}$$

Its finite commutators connect source current, background change, and matter force, yielding the Maxwell response in a weak-phase long-wavelength window. Color fields retain the complete Gram operator of ordered matrix loops. Massive mediators retain their actual endpoint numerators. Pion exchange combines complete nucleon action with odd spatial arms to produce spin-, flavor-, and direction-dependent attraction and repulsion. Physical identifications are inherited, while explicit joint constructions establish the mechanisms with their parameter calibration and approximation regimes retained.

The finite light-excitation process connects source lowering, background generation, propagation, and energy balance. The conditional directional-radiation construction likewise preserves constraints and gives reciprocal absorption. The full background, however, does not contain only two modes. Connecting its new amplitude to the original material clocks and rods still requires constraint-preserving formation and readout. Relative electromagnetic-return calibration, control of persistent strong-background correlations, and this material connection delimit the present results from a more complete field theory.

Field observation continues within these same dynamics. Changes in probe motion, phase, and clock rate leave distinguishable records through the apparatus; the joint final state Ω^{out}, including the detection process, and the pointer projection Π_r give the event probability

$$p_r = \mathrm{Tr}\left[(I \otimes \Pi_r)\Omega^{\mathrm{out}}\right].$$

While transferring energy and momentum, light connects distant signals and clock comparisons; an interferometer converts phase differences into outcome probabilities, and local absorption converts energy exchange into counts. Event positions and times are represented using a common three-dimensional space and one-dimensional time. When statistical readings are stable, spatial variations are smooth, and correlation contributions are controlled, these responses support a continuum field description. Matter produces fields and manifests their action within the same evolution, giving field quantities and their observation a common dynamical origin.

Data and Computation Statement

This paper uses no new experimental or observational data; numerical calculations are used to check analytical results and produce figures.

Competing Interests

The author declares no competing interests.

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